Option Value and Storefront Vacancy in New York City*

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Abstract

Why do storefronts remain empty for more than a year in some of the world's highest-rent retail districts? Landlords with vacancies derive option value from two sources of uncertainty. First, increasing downstream retail demand may drive up market rents tomorrow. Second, different tenants may have different willingness to pay for the same space, creating an incentive for landlords to wait for a particularly high rent offer. Search frictions, move-in costs, and eviction costs amplify this option value. This paper builds and estimates a two-sided, dynamic structural model of the commercial leasing market using high-frequency data on the universe of Manhattan storefronts and a sample of lease contracts. In New York, we find that option value deriving from variation in match quality drives the long-run vacancy rate: reducing the variance of the match quality distribution by 50% reduces long-run vacancy rates by 33% on average, while reducing the variance of the aggregate state variable has almost no effect. Finally, we find that a proposed vacancy tax would incentivize landlords to shorten vacancies by leasing to lower-quality tenants, who would also be more likely to exit early. As a result, the tax has only a small impact on vacancy rates and market rents. The welfare-maximizing vacancy tax in the absence of externalities is close to zero in most neighborhoods. Vacancies would have to generate negative externalities of $18.72 per square foot per quarter (about 30% of average rents) to justify a 1% vacancy tax on assessed property values.

1 Introduction

Persistently vacant storefronts, especially in the dense pedestrian retail corridors of cities like New York, San Francisco, or Chicago, often elicit vociferous complaints from local communities. Pedes-

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trians and residents may lament the "waste of space" which could be used to provide local amenities, worry that less economic activity on a block will lead to higher crime rates (Jacobs, 1961), or simply feel that vacancies reduce curb appeal and neighborhood atmosphere (Glaeser et al., 2018; Freybote et al., 2016). While short-term vacancies may impose a fleeting cost associated with tenant turnover, long-term vacancies in high-rent urban areas pose more of an economic puzzle: why do landlords in dense urban areas appear to forego high rents and leave their retail properties vacant for a year or longer?

We study this puzzle in the context of the Manhattan retail space market prior to the COVID-19 pandemic, where long-term storefront vacancies are the norm rather than the exception. Figure 1 displays the CDF of vacancy distributions in our data. Strikingly, only 50% of vacancies were filled after one year, and the average vacancy spell between 2015 and 2019 lasted 16 months. These long-duration vacancies were common despite the fact that annual rents average $240 per square foot during our study period (often topping global retail rent rankings).

Why would landlords choose vacancy when market rents are so high? The fact that landlords choose to forgo high market rents suggests that demand for real estate space is robust, and instead that vacancy is a choice landlords make to maximize long-run profits. In this paper, we investigate two sources of option value that induce landlords to wait before filling vacant storefronts: aggregate time series uncertainty in market rents and cross-sectional heterogeneity in tenant willingness to pay. Time series uncertainty creates an incentive to sign rents as close to the top of the rent cycle as possible. If a landlord thinks market rents will increase next quarter, perhaps due to an increase in downstream demand for retail goods and services, they will choose to wait rather than signing a lease today. Cross-sectional heterogeneity in tenant willingness to pay for space – which is reflected in substantial contractual rent variation – creates an incentive for the landlord to wait for a tenant offering above-market rents.

We attribute heterogeneity in tenant willingness to pay to match quality between the landlord and tenant. Some storefronts are better suited to some types of retail activity than others: tenants' space needs, in terms of quantity as well as other characteristics, range widely across tenants, both across and within industries. Coffee shops, bank branches, restaurants, and clothing stores all have different space needs from one another, but Starbucks is also likely to have different space needs from a local coffee shop. Search and matching frictions in the leasing market make it difficult for landlords to inspect every possible tenant (or for tenants to inspect every possible space) immediately upon becoming vacant. As a result, a landlord has an incentive to wait and search for a particularly well-matched tenant, who might be willing to pay above-average rents for their particular space.

In this paper, high quality data on storefront vacancy and rents allows us to characterize features of Manhattan's real estate market which amplify landlords' option value from vacancy. First, 58% of leases have 10-year terms, and the vast majority feature non-state-dependent nominal rents. As a result of this lease structure and tenant protection laws which make eviction costly, landlords are locked into leases with real rents which are likely to decline over the course of the lease. This provides a strong incentive for landlords to lock in high rents up front. Second, most tenants exit before their lease expires. While this means that landlords can expect to reset their rents more frequently than every 10 years, tenants are more likely to exit when downstream retail demand (and therefore market rents) are low. This creates an incentive for landlords to find a tenant who is a "good match" and therefore less likely to exit in response to negative aggregate retail demand shocks. Third, rents vary widely both within and across neighborhoods. About 62% of this variation can be explained using observable landlord and tenant characteristics; we attribute the remainder to landlord-tenant match quality.

We then construct and estimate a dynamic search-and-matching model which takes these institutional details of the retail real estate leasing market into account. The model builds off of the industrial organization literature on decentralized, two-sided matching models, incorporating both landlords' two sources of option value: aggregate uncertainty and heterogeneous match quality. Though we estimate the model using data from Manhattan retail markets, it can be broadly applied to leasing markets in different geographic and space type markets[1].

Our estimated model allows us to quantify the extent to which our two sources of option value– aggregate uncertainty and match quality– contribute to long-run vacancy rates. In a counterfactual exercise, we scale down the variances of uncertainty in downstream demand and heterogeneity in match quality. These variances are the sources of option value – if there was no variation in downstream demand over time or in the match quality associated with new tenants arriving each period, landlords would always accept the first tenant to arrive and there would be no incentive to

[1] For example, it could easily be adapted to study office markets in Chicago or industrial leasing markets in California's Inland Empire

wait. We find that heterogeneous match quality is the primary driver of vacancy rates over long time horizons: scaling down the variance of the match quality distribution by 50% reduces long-run vacancy rates by 33% on average, while reducing the variance of the aggregate state variable has almost no effect on long run average vacancy rates.

Finally, we use the model to assess the impact of a proposed vacancy tax on the leasing market, and in particular on long-run vacancy rates. As of April 2025, a storefront vacancy tax is currently under consideration in the New York State legislature.[2] Proponents of the tax argue that landlords who keep storefronts vacant depress local economic activity and pose a threat to neighborhood safety by reducing the number of "eyes on the street" (Jacobs, 1961). They view the tax as a Pigouvian measure which would cause landlords to internalize the impacts of their vacancies on urban vibrancy. Retail vacancy taxes of this nature have been implemented over the last decade in Washington D.C., San Francisco, and Oakland, California. We find that New York's proposed commercial vacancy tax would indeed encourage landlords to fill vacant spaces more quickly, reducing vacancy rates and retail rents. However, the tax would also distort the set of tenants in the market, with lower-earning retail stores arriving at opportune moments crowding out higher-earning stores that might have arrived later. These lower-earning stores are more likely to exit, increasing retail churn and reducing welfare. We find that, in order to justify the proposed tax, each vacant square foot would have to generate a negative externality of $18.72. This represents about 30% of average rents, and is probably larger than the actual externality associated with retail vacancy.

Related Literature. This paper relates most closely to the literature on commercial real estate leasing markets. The total value of US commercial real estate was roughly $18 trillion in 2018 (NAREIT, 2019); the value of these assets depends directly on the rents landlords are able to command, yet the forces that shape occupancy rates and rents have received relatively little research attention. CRE in general, and the office sector in particular, has attracted renewed interest following COVID-19, with recent work studying how office markets responded to remote work (Gupta et al., 2025) and how lease expirations affected property performance (Glancy and Wang, 2025). Within retail, the literature has focused more on mall leasing than on high street leasing (Konishi and Sandfort, 2003; Benjamin et al., 1992; Brueckner, 1993; Burayidi and Yoo, 2021), in part because the economics are quite different: mall operators internalize the externalities that

[2] Senate Bill S6804/Assembly Bill A669. See https://www.nysenate.gov/legislation/bills/2025/S6804.

tenants impose on one another, whereas storefront ownership in high street markets is fragmented. Brooks and Meltzer (2025) find that zoning codes may drive an oversupply of downtown retail space, creating excess vacancy. If so, the appropriate policy response may be rezoning rather than a vacancy tax. We contribute a dynamic, 2-sided model of retail leasing in a downtown or high street context, and focus on the mechanisms generating vacancy in normal times. While we estimate the model using data from Manhattan's retail market, the framework could be applied to other CRE asset classes and geographic markets.

Because signing a long-term lease is, for a landlord, a temporarily irreversible investment under uncertainty, our work connects to the literature on real options and investment decisions in real estate contexts. Capozza and Helsley (1990) show that uncertainty affects equilibrium land prices even when landowners are risk neutral. There is a long literature quantiying option value effects in real estate development (e.g., Bulan et al., 2009; Clapp et al., 2013). Within leasing markets specifically, Grenadier (1995) and Ambrose et al. (2002) derive models in which fluctuations in market conditions create option value. We contribute to this literature by taking the analysis to a retail leasing context and considering option value arising not only from aggregate market uncertainty but also from idiosyncratic landlord-tenant match quality. Furthermore, we estimate the parameters driving both sources using data from Manhattan's retail market.

Finally, our paper connects to a literature documenting how retail amenity changes shape neighborhoods. This literature shows that small demand shocks can have large effects through endogenous retail responses, in the context of neighborhood revitalization (Caplin and Leahy, 1998), gentrification and re-urbanization (Couture and Handbury, 2020; Couture et al., 2021; Su, 2022), e-commerce (Quan and Williams, 2018; Tran, 2021), and the COVID-19 pandemic (Duguid et al., 2024). While these papers typically focus on aggregate neighborhood-level outcomes, we focus on the intermediate market that must clear before any retail service reaches final consumers: the leasing market between landlords and tenants. We also connect to the literature quantifying vacancy externalities in residential (Anenberg and Kung, 2014) and commercial real estate contexts—higher crime rates (Jacobs, 1961; Chang and Jacobson, 2017; Humphrey et al., 2020), reduced foot traffic (Pashigian and Gould, 1998; Shoag and Veuger, 2018), and lower surrounding property values (Billings et al., 2024)—which provides the normative motivation for the vacancy tax we evaluate.[3]

[3]

The paper proceeds as follows. Section 2 describes our data. Section 3 documents the institutional context we build into our model, which we present in section 4. Section 5 estimates the model and reports the estimated parameters. In section 6, we quantify the relative strengths of the frictions in our model by counterfactually shutting them down one at a time and looking at the effect on long-run vacancy rates. Section 7 performs the counterfactual vacancy tax exercise. Section 8 concludes.

2 Data

Commercial leasing markets (especially retail leasing markets) are understudied largely as a result of a lack of high-quality microdata. Our analysis relies on two key datasets from commercial real estate data companies which provide high-quality storefront- and lease-level data. One dataset is a novel panel of storefront-level occupany and vacancy covering the universe of storefronts in Manhattan from 2015-2019. The other is a sample of lease contracts with detailed information on contractual rents, lease start and end dates, and tenant identities. We also use a city government report on retail vacancy rates to extend our vacancy data back to 2007, and aggregate data on retail sales to capture uncertainty in downstream retail demand.

Our datasets represent the best available information on retail vacancy and contractual rents at the storefront and lease levels, respectively. While city governments collect information on occupancy rates and rent rolls in confidential tax filings, this information publicly available.[4] Commercial real estate brokerage firms publish semi-annual reports on high-rent retail markets, but these reports are not ideal for academic research for several reasons. First, they report aggregated data exclusively from highly selected retail corridors rather than on the city as a whole. These retail corridors are the most desirable areas to place a retail store - they are located in high-traffic areas close to public transit stations, and buildings (and their retail spaces) tend to be larger. Second, broker's reports measure slightly different objects than those we are interested in for the purposes of this study. Specifically, they document asking rather than effective rents[5], and track storefront availability (spaces which landlords have listed for rent) rather than vacancy (which is usually the

[4]New York has collected data on vacancy rates from tax filings since at least 2007, but did not make this information public until 2019. 2019 is also the year the city began its public storefront registry in 2019 under Local Law 157.

[5]In our data, net effective rents can be as low as 60% of asking rents.

quantity of interest to pedestrians, residents, and policymakers)[6] Finally, while retail real estate multiple listing services provide information on asking rents and time on the market, they also measure availability rather than vacancy and do not track occupied spaces over time.

Market Definition We report all summary statistics and estimate our model at the neighborhood level. Our 8 neighborhoods correspond to Manhattan community districts 1 through 8. Community districts are large, geographically contiguous neighborhoods that are represented by community boards. New York City has 59 Community Districts across all 5 boroughs, ranging in population from 50,000 to 200,000 residents. We focus on the 8 community districts in Manhattan south of 110th street, since these are the areas in which we have sufficient occupancy and leasing data to compute our moments each quarter. While our 8 community districts do not cover the whole city, they do make up the areas with the densest and most valuable retail space.[7]

2.1 Live XYZ vacancy and occupancy panel

Our first dataset is a novel storefront-level panel of occupancy and vacancy for the near-universe of Manhattan retail storefronts between 2016 and 2019. These data come from a firm called Live XYZ, which employed "mappers" to track stores and storefronts in person and over the phone. They tracked traditional goods retailers (NAICS codes 44-45) as well as restaurants and bars (NAICS code 722), and service providers (NAICS code 812).[8] Mappers traveled around the city, mapping the GPS location of each storefront's door and assigning individual storefront identifiers to each. Over time, they tracked information about each storefront's "status" - whether it was vacant, occupied, or under construction, the type of tenant if there was one, and the store's hours of operation. They also verified store status by calling retail businesses. Live XYZ stopped operating shortly after the onset of the COVID-19 pandemic.

The Live XYZ dataset is key to our study of retail vacancy because data on neighborhood retail vacancy rates is generally not publicly available. Many local governments do not systematically

[6]The availability rate tends to be much higher than the actual vacancy rate because it includes occupied spaces where the lease is going to expire soon or the tenant intends to exit and wants to sublease the space. The availability rate can be lower than the vacancy rate, however, because it does not include stores which are temporarily closed or storefronts which have been abandoned.

[7]Estimating the model at a more spatially disaggregated level is theoretically possible, but the number of leases we observe per market in each quarter begins to rapidly decline as we split the markets into smaller and smaller neighborhoods.

[8]In this paper, we will therefore use the term "retail" to refer to all three types of stores.

collect vacancy data or (as in New York City's case) do not make it available to the public. Real estate associations such as the Real Estate Board of New York publish quarterly market reports, but these tend to focus on highly selected retail corridors (for example, Fifth Avenue between 42nd Street and 49th Street) and often report "availability" rather than vacancy. Traditional retail establishment panels (such as Dun and Bradstreet, Infogroup, Yelp, or Google Reviews) used in many papers do not record vacancies directly and would have required us to infer vacancy from a lack of data on occupancy at the suite (sub-address) level. By contrast, our dataset provides consistent suite or storefront-level identifiers over time, which means that we do not have to rely on messy sub-address level matching to track storefronts over time or infer that a storefront is vacant when data on a given store is not included.

The Live XYZ panel allows us to compute the vacancy rate, the lease-up rate (the share of vacant spaces which become occupied) and the tenant exit rate (the share of occupied spaces which become vacant) for each neighborhood in each period. These are all key moments we will attempt to match when we estimate our structural model. The dataset tracks detailed information about each storefront and its occupant (or lack thereof) over time, allowing us to construct a panel on storefront vacancy and occupancy at the quarterly level. Persistent storefront-level attributes include its address and the precise GPS coordinates of the door. Time-varying attributes include an indicator for whether the storefront is occupied, under construction, or vacant; the name of the tenant that occupies it, if there is one; and whether or not the tenant is operating, coming soon, closing soon, temporarily closed, or permanently closed. Each state is labeled with its start and end date. The dataset also records the industry of the tenant (for example, whether it is a restaurant or an apparel store), subcategory (for example, the type of cuisine a restaurant serves), and its parent chain if it has one. In addition to traditional goods retail tenants, we also observe all other first-floor business tenants, including restaurants, bars, dry cleaners, personal care salons, bank branches, and private medical practices, among many other establishment types. Live XYZ tracks changes in a storefront's state by scraping individual store websites and Facebook pages, calling stores directly, and physically visiting store locations.

The main limitation of the Live XYZ dataset is that it covers a relatively short period of time: the end of 2016 through February 2020.[9] Therefore, most of the tenants we observe are either

[9] The dataset actually begins in 2015, but rapidly adds observations through the end of 2016.

installed in a space already by the time the dataset starts, are still in the space when the dataset ends, or both. To avoid selection bias as the dataset is build up, we only use Live XYZ to compute neighborhood-level vacancy rates beginning in 2017Q1.

We report summary statistics on this dataset in table 1. We observe a total of 21,811 storefronts across the 8 community districts we estimate the model on, and the average vacancy rate across quarters and neighborhoods is 5.23%.

2.2 New York City Comptroller's Report

We augment the Live XYZ occupancy data using zip code level data on retail vacancy rates from a city report on retail vacancy (Office of the New York City Comptroller, 2019). The report provides vacancy rates for each borough over the 2007-2017 period, as calculated from landlords' annual property tax filings. It provides the same statistic for select (but not all) zip codes, which we aggregate to the community district level.

2.3 Lease contract data

We combine our occupancy data with micro data on lease contracts from CompStak. These data allow us to observe contractual rents for different retail spaces and in different time periods. In addition to contractual rents, we observe a lease's execution date (when the lease is signed), commencement date (when the tenant moves in) and expiration date, the identity of the tenant, the address of the property, and physical property characteristics such as the size of the space (in square feet). Our CompStak sample contains leases executed between 2005 and 2019.

The CompStak dataset gives us lease-level information for a broader sample of rents in a neighborhood than data reported by commercial real estate (CRE) agencies. CRE agencies often publish aggregated quarterly or semi-annual reports on the state of the leasing market, but they focus on relatively small and highly selected retail corridors and usually report only average rents for those areas. The CompStak data contain leases from the retail corridors that CRE brokers report on, but also contain leases for retail stores throughout the city. CRE agencies also typically do not report on other contractual provisions, including commencement and expiration dates, tenant improvements, or other concessions. This information allows us to compute net effective rents, which are more relevant for tenants' and landlords' decision-making than asking rents.

Rents are almost always quoted in nominal dollars per square foot, and are either constant over the lease term or include fixed step-ups at contractually specified dates. The main rent variable we use in our analysis is "net effective rent", which factors in both contractual rent increases and concessions the landlord grants to the tenant. Where reported, rent step-ups usually occur every 1 to 3 years, and are usually an increase in rent of a few percentage points. Landlord concessions take the form of either tenant improvements (payments the landlord agrees to make to help the tenant renovate the space) or months of free rent (time at the beginning of the lease when the tenant occupies the space without making rent payments). Leases are usually executed the quarter before commencement.

CompStak crowdsources lease information from commercial real estate brokers, which means our dataset is likely to reflect a selected sample of properties. Brokers are incentivized to share details from contracts they were involved with because doing so allows them to access more lease comparables themselves. This means our sample contains lease information primarily about deals that brokers were involved with and which are not so sensitive that brokers are unwilling to share information about them. Though we do not have data on the volume of non-brokered deals in New York, conversations with industry participants suggest the majority of lease transactions are mediated by brokers. The crowdsourced nature of the CompStak dataset also means that our rent observations are likely to contain measurement error. For example, brokers do not always report the full contractual rent schedule or the lease concessions. However, we think this measurement error is negligible, since the magnitude of step-ups and concessions are generally small when compared to the overall value of the lease.

From the lease-level CompStak microdata, we compute the mean and standard deviation of net effective rents for leases executed in each geographic submarket in each quarter. When we estimate our structural model, we will treat these time series as moments to match. Information on the spread of rents is important because it helps us identify the degree of heterogeneity in tenants' willingness to pay for retail space, which is a key parameter in our model.

Table 1 reports summary statistics from this dataset alongside the summary statistics for the Live XYZ dataset. We have a sample of 7,991 leases for properties located in our 8 community districts, all of which were executed between 2005 and 2019. Given that the CompStak data is a sample of leases while Live XYZ covers the near-universe of storefronts, we observe many fewer leases than storefronts in each neighborhood. We can see that some neighborhoods are better represented in our dataset than others: in particular, we observe the most leases in the three high-end retail districts: Midtown (which contains Times Square and Fifth Avenue), the Lower West Side (which contains SoHo and the Village), and the Upper East Side (which contains Madison Avenue). There is also some selection over time. Figure 4 shows the number of leases executed each quarter. CompStak itself entered in 2012, and the size of their dataset grows over time.

Table 1 shows that there is a negative correlation between average rent and vacancy rates. For example, the neighborhood with the highest average vacancy rate and the lowest average rent is the Lower East Side, while the Upper East Side has the highest rents and the second-lowest retail vacancy rate. One might be tempted to conclude from this correlation that more desirable neighborhoods draw more demand for retail space, and this bids up prices and reduces vacancies (i.e. increases the quantity of occupied space). However, because rents (prices) and vacancy rates (quantities) are jointly determined in equilibrium, we cannot interpret this correlation as causal. The role of our structural model is to use variation in rents and vacancy rates across time and within markets to identify key structural parameters of the model, which we can then use to make counterfactual predictions about the effects of policy changes on rents and vacancy rates.

3 Institutional Detail

In this section, we document three main features of the commercial real estate leasing market that contribute to retail vacancy and will inform our structural model in section 4. First, retail lease terms are long: 58% of leases in our data have 10-year terms. Retail lease terms are among the longest across all types of CRE, primarily because of the high costs tenants incur to renovate their spaces upon move in. Long lease terms protect tenants from being held up for high rents before these move-in costs have been recouped.

Second, despite these long lease terms, 54% of tenants exit before their lease expires. In New York, tenants can unilaterally exit leases more easily than landlords. This creates option value for the two parties at different times during their relationship: landlords have option value while vacant, while tenants have option value while the lease is in effect.

Finally, there is substantial rent dispersion both across and within neighborhoods, even when controling for tenant and landlord characteristics, and there is a long right tail in retail rents. This rent dispersion suggests that there is substantial unobserved heterogeneity in tenants' willingness to pay, and that search frictions play an important role in this market.

3.1 Lease terms are long

The retail real estate market is characterized by very long-term leases, even relative to other commercial real estate markets. Figure 6 shows a histogram of lease lengths in our retail lease dataset. In our sample, 58% of leases have a contractual term of exactly 10 years. The second most common lease length is 15 years (accounting for about 12% of leases in our data), followed by 5 years (accounting for about 9% of leases). By comparison, most residential leases carry one year terms, and office leases usually have terms of five to seven years.

Retail leases carry long contractual terms primarily to allow time for tenants to recoup high up-front move in costs. Storefront appearance is a key part of a tenant's visual brand, so retailers are willing to invest in custom build-outs that help convey that image. This is especially the case in New York's high-rent retail districts (such as Madison Avenue, Fifth Avenue, Times Square and SoHo) where large chains place their flagship establishments. Move-in costs can also include items other than renovations, such as specialized equipment (restaurants often rent their large appliances), permit fees, hiring costs, and advertising costs. Unfortunately, lease-level data on move-in costs is hard to obtain. We will calibrate them in our structural model.

There are also reasons to believe that long lease terms are attractive to landlords. Long-term leases provide evidence that landlords will have stable cash flows over the medium term, which may make it easier for them to borrow.

3.2 Nearly 55% of tenants exit before their lease expires

Although lease terms are long, most tenants actually exit prior to their lease's expiration date. We estimate that 55% of tenants on 10-year leases have exited within 5 years of signing a lease. This, along with the fact that formal evictions are rare, suggests that tenants have substantial discretion over whether or not to exit their lease early.

Because CompStak does not track landlords and tenants after leases are signed, we estimate tenants' age at exit by matching leases to stores observed in the Live XYZ occupancy data. We are able to match 3,248 leases to stores in our occupancy dataset. In this matched sample, 2,107 leases have 10-year terms. Among these, 462 of the tenants exit at some point during Live XYZ's four years of observation.[10]

Figure 7a plots a histogram of the tenant's lease age at the time of exit, based on our matched sample. Only about ten percent of the tenants in this sample actually exit at the lease expiration date. A very small sample of tenants (composing less than 5% of the matched sample) appear to have renewed their leases, since we see them exit after their contractual lease term ends. By far the majority of tenants exit early. Exit is more common at the beginning of the lease; the most common age at exit is 2 to 2.5 years. Figure 7b shows that 20% of tenants have exited after two years, and 54.8% of tenants have exited after five years.

Tenants are able to exit early at low cost because of a standard lease clause in New York City retail leases called the "good guy guarantee." This provision allows the tenant to provide the landlord with advance notice that they will leave the space on a given date (usually 3 months in advance), called the surrender date. The tenant then pays all their rent obligations to the landlord through the surrender date, and vacates the space. All rent obligations after the surrender date are then cancelled. This clause is common in New York because it benefits both parties: the tenant gains limited liability in the case of bankruptcy, while the landlord does not have to go through the onerous eviction process and is able to fill the space again more quickly. Of course, the landlord does face a cost when the tenant exits (he stops earning rents, and has to find a new tenant to fill the space), but the ubiquity of the good guy guarantee in New York suggests that on net landlords find it worthwhile to provide tenants with a cheap exit option rather than risk needing to evict a nonpaying tenant.[11]

By contrast, it is costly in time and money for landlords to unilaterally exit a lease. They cannot remove a paying tenant just because market rents have changed, or because another potential tenant offers to pay a higher rent. Even if their tenant has broken the terms of the lease (by defaulting on

[10] CompStak dataset only gives us contractual lease length and not the ex post amount of time the store stays in the space. To determine ex post duration, we match leases to stores in the Live XYZ occupancy dataset, using name, address, and dates of occupancy.

[11] The "good guy guarantee" appears to be a specific feature of New York City real estate markets, but Mooradian and Yang (2002) and Yoshida et al. (2016) both show evidence that tenants pay premiums for short-term leases or cancellation options in other contexts. In general, leases often specify a "guarantee" which tenants must pay if they choose to exit a lease early. We do not observe whether contracts contain the "good guy guarantee" and so cannot identify the rent premium associated with this cancellation option, but assume that it is priced in to the average lease.

rent or breaking other lease covenants), eviction is costly.[12] This inability to unilaterally dissolve an ongoing lease incentivizes landlords to be very selective when signing leases in the first place. It is often worth it for landlords to remain vacant for several quarters if there is a good chance that a high-paying tenant may come along later. The value of remaining vacant that results from long lease terms and landlords' inability to unilaterally exit from existing leases is what we will refer to as "landlord option value" for the remainder of the paper.

Because tenants can exit leases unilaterally at any time during the lease term while landlords are more constrained, tenants and landlords have option value at different points in their contractual relationship: the tenant has option value during the lease (since they can always choose to exit), while the landlord has option value while vacant. Because landlords cannot exit a lease whenever they choose, signing a lease is similar to making an irreversible investment under uncertainty (Dixit and Pindyck, 1994). A central premise of the real options literature is that there is option value in waiting for new information to arrive, so capital owners often delay investment and/or require compensation for that option value in the form of higher rental rates. Bulan et al. (2009) show this mechanism is quantitatively important in real estate development; our model will illustrate a similar mechanism at work in leasing markets for existing buildings.

3.3 There is substantial rent dispersion within and across neighborhoods

Rent dispersion across New York City retail leases is large, and cannot all be explained by observable characteristics of landlords or tenants. Figure 5 shows the histogram of real rents for each neighborhood in our sample, pooled across all periods. It shows that, in all of our neighborhoods, rents have a long right tail. While most tenants pay between \$10 and \$50 per square foot, in most neighborhood there is a small number who pay upwards of \$75 or even \$100 per square foot.

We use our matched sample of tenants to run a hedonic regression of rents on observable tenant and landlord characteristics. Table 2 shows the results of this regression. Column (1) includes transaction quarter fixed effects, column (2) adds tenant industry fixed effects at the 3-digit NAICS level, column (3) adds zoning fixed effects, and column (4) adds census tract fixed effects. As expected, statistical significance of the correlation between building-related characteristics and rents

[12] To evict an unwilling tenant, landlords must go to court and show the tenant has broken the terms of the lease. The eviction process takes a long time, and legal fees are expensive.

mostly vanish once zoning and census tract fixed effects are added. The most persistently significant coefficients are on the dummy for whether a store opens on to an avenue (a large thoroughfare running north-south, rather than the smaller east-west street) and on whether the tenant is a chain store. The term premium is statistically significant, which we expect to see because longer lease lengths increases tenants' option value. Finally, we find that there is a significant quantity discount (tenants who rent more square feet of space pay less per square foot).

Even when we saturate the regression with fixed effects, these observable characteristics account for 62% of rent variation overall and only 15% of rent variation in each cell. In other words, there are many instances of observably similar tenants paying very different rents for similar spaces in the same neighborhood. Our main takeaway from the extent of rent dispersion is that pricing does not appear to be competitive, and our model will need to assume a different pricing mechanism to be able to generate this rent dispersion.[13]

In our structural model, we will explain residual rent dispersion with search frictions and match quality heterogeneity.[14] Real estate markets in general are characterized by high search frictions. These are markets in which both landlords and tenants are heterogeneous, and finding a "good match" can be challenging.[15] In this paper, we assume that search frictions force landlords and tenants to encounter each other sequentially, so that their choices can be approximated by optimal stopping problems.[16] Heterogeneity across landlords and tenants also plays an imporant role and realistic: different tenants have different willingness to pay for the same space, depending on their business model, target customer base, and other factors. Without such heterogeneity, there would be no reason for landlords to leave spaces vacant, since they could always find a tenant willing to pay the market-clearing rent.

[13]Stanton and Wallace (2009) show that standard no-arbitrage pricing models cannot account for observed rent dispersion across commercial property types.

[14]Deng et al. (2012) study the equilibrium effects of housing price dispersion on asking prices, using a theoretical model similar to the structural model we will posit. We do not attempt to model this channel in this paper, but note it as a possible area for future research.

[15]Indeed, the real estate brokerage industry exists to help ameliorate these search frictions, and makes substantial profits doing so.

[16]Han and Strange (2015) reviews several models of search and matching models in housing markets.

4 Model

In this section, we present a dynamic model of the storefront leasing market which incorporates the market features described in section 3: search frictions, move-in costs, match quality heterogeneity, endogenous tenant exit, and aggregate uncertainty. After estimating the model in section 5, we quantify the relative importance of our two sources of option value in section 6. We will evaluate the consequences of a counterfactual vacancy tax in section 7.

Our model builds upon recent models of search and mtaching in the industrial organization literature, especially Vreugdenhil (2020) and Brancaccio et al. (2020b). Vreugdenhil specifies a two-sided search and matching model with observable heterogeneity on both sides of the market, and seeks to explain a pro-cyclical assortative matching pattern. In our context, match quality is not a straightforward function of landlord and tenant characteristics, and we (the econometricians) cannot observe it. Our goal is to identify the distribution of match quality by imposing sufficient structure on observed rents, lease-up rates, and exit rates. Our model also can explain endogenous early contract dissolution, which (as we documented in the previous section) is an important feature of the commercial real estate market that is not observed in other markets in which search-and-matching models of this type are usually estimated. In the context of bulk shipping (Brancaccio et al., 2020b) or taxi riding (Buchholz, 2022; Fréchette et al., 2019), the duration of relationships between counterparties is determined by the (exogenous) amount of time it takes to travel from one place to another. Labor market search-and-matching models often assume that job arrangements are at will.[17] Our model of endogenous contract dissolution is relevant for other settings with long contract terms where market participants cannot commit to a contract. This is a prominent feature of markets with prepayment risk, such as life insurance markets (Hendel and Lizzeri, 2003) as well as many consumer lending markets (for example, residential mortgages and auto loans).

4.1 Environment

We model the endogenous formation and dissolution of leases between landlords and tenants within a neighborhood (which we refer to as a "market"). Each tenant needs only a single storefront to operate their business. Time is discrete and infinite-horizon. There is no information asymmetry

[17]The canonical labor market search-and-matching models are Mortensen and Pissarides (1994) and Hosios (1990).

between landlords and tenants, and all leases have a contractual term of T periods.[18] We assume that the set of storefronts is fixed.[19] Within a neighborhood, all landlords and tenants face the same distribution of possible match qualities, but we allow the distribution of match quality to vary across neighborhoods.

Figure 1 illustrates the sequence of moves in each period. First, the aggregate state of downstream retail demand g_t is drawn from a known stochastic process and observed by all agents. The realization of the aggregate state affects all tenants' gross profits, but does not affect landlords' payoffs directly. Tenants also draw an idiosyncratic opportunity cost ϕ_{it} each period, which affects their decision to continue operating or exit the market. If the tenant chooses to continue operating, they earn gross profits (a function of the aggregate state and their match quality with the landlord), pay rent, and continue to the next period. If the tenant chooses to exit, they cease operating, and earn their opportunity cost. Exiting tenants remain obligated to pay the current period's rent under the good guy guarantee. When a tenant's lease ends, either by choice or by reaching the lease expiration date, the tenant exits forever.

At the beginning of each period, each landlord l is either vacant or has an incumbent tenant i. When the landlord is vacant, the lease is in its final period before expiration, or i has invoked the good guy guarantee, the landlord draws a new potential tenant n with some probability. If the landlord successfully draws or "matches with" a potential tenant n, they draw their match quality with n, θ_{ln}, from a fixed, exogenous distribution of potential tenants. Observing θ_{ln}, the landlord either makes n a take-it-or-leave-it rent offer for a lease of fixed length T, or rejects them (choosing to remain vacant and search again the next period). If n accepts the rent offer, they move in, sink move-in costs and begin paying rent in period $t+1$.

The idiosyncratic state variable of a landlord includes its contractual rent due this period, r, and the age of its current lease, j. If the landlord is vacant, then $j_{lt} = \text{vacant}$ and $r_{lt} = 0$.

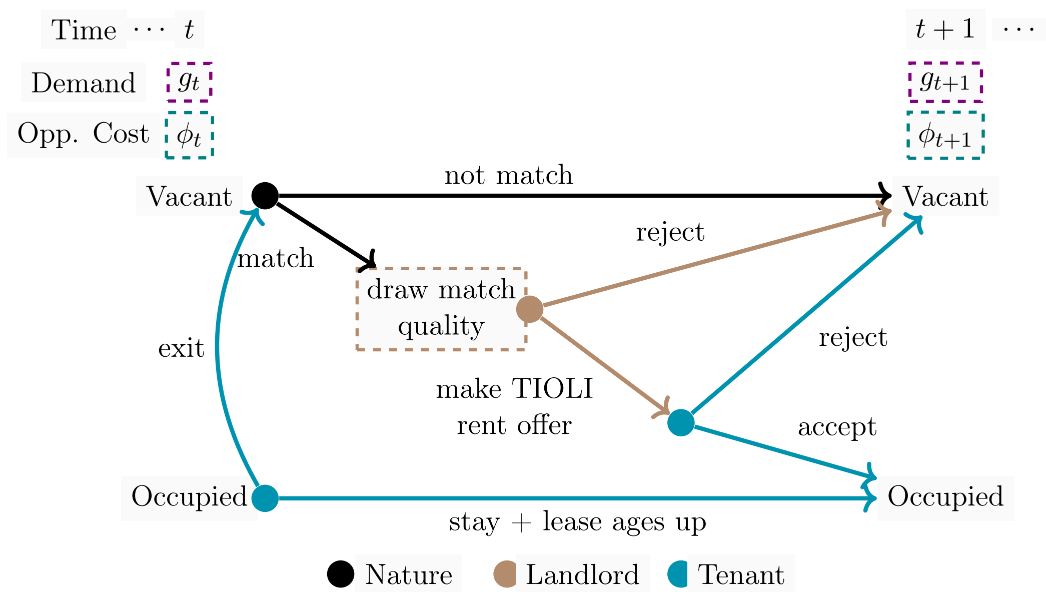
Downstream retail demand follows an AR(1) process:[20]

[18] In reality, the length of a retail lease is a negotiable contract feature. However, 10-year leases are by far the most common in the data. To simplify our model, we assume that all leases have fixed 10-year terms. We could relax the model to allow a menu of possible contract terms, but this would substantially increase the computational burden to solve the model and most tenants exit prior to the end of the lease term anyway.

[19] Davidoff (2010) notes that there is almost no vacant land left in Manhattan. Though existing buildings could be redeveloped, especially if zoning restrictions were relaxed, we abstract from entry and exit of storefronts in this model.

[20] Note that this AR(1) process can be re-written as $g_t = \delta + \rho_g g_{t-1} + \varepsilon_t$ where $\mu_g = \frac{\delta}{1-\rho_g}$. We choose to express

Figure 1: Agent choices flow chart



This flow chart illustrates the interdependence of landlords' and tenants' choices in the model. Black lines and dots denote moves made by Nature. Brown dot and lines denote landlords' decision points and choices, respectively. Blue dots and lines denote tenants' decision points and choices, respectively. First, the aggregate state g_t is realized and observed by all agents. If the landlord is vacant, they search and match with a potential tenant with some probability. If they match with a potential tenant, they draw a match quality and either make a take-it-or-leave-it rent offer or reject the tenant. If they reject the tenant, they enter the next period vacant and repeat the process again at time $t+1$. If the potential tenant accepts the rent offer, they move in and the landlord becomes occupied at time $t+1$. If the landlord enters the period occupied, the tenant chooses whether to continue or exit. If they continue, the landlord has no choices to make. If the tenant exits, the landlord behaves as a vacant landlord would. The dashed boxes indicate the places where uncertainty is resolved. The purple boxes indicate times when aggregate uncertainty is resolved (the realization of g_t in period t is common to all agents, and is observed by all at the beginning of the period). The brown box highlights the time when uncertainty over match quality, which is idiosyncratic to each landlord-tenant pair and which the landlord can act upon unilaterally, is resolved. The blue boxes indicate the tenant's opportunity cost of continuing to operate, which is idiosyncratic to each tenant in each period. Tenants can respond to their opportunity cost realizations unilaterally.

$$g_t - \mu_g = \rho_g(g_{t-1} - \mu_g) + \varepsilon_t \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_g^2) \tag{1}$$

4.2 Tenant Payoffs and Behavior

Tenant flow payoffs Each landlord-tenant pair li draws a match quality θ_{li} which is drawn from an exogenous distribution F^θ after the tenant enters the market. θ_{li} is fixed for the duration of the pair's contractual relationship. For estimation, we assume F^θ is lognormal with parameters μ_θ and σ_θ.

Let $s_{it} = (j_{it}, r_{it}, g_t)$ denote tenant i's state in period t. j_{it} is the tenant's lease age in period t, r_{it} is the rent the tenant owes in period t, and g_t is the aggregate state in period t. j_{it} and r_{it} evolve deterministically: rent is simply fixed over the lease's term and j_{it} increases by 1 in period $t+1$ if the tenant chooses not to exit in period t.[21]

Per-period tenant gross profits are a multiplicative function of the aggregate tenant profitability state g_t, and match quality θ_{li}. The deterministic portion of gross profits is given by

$$\pi(g_t; \theta_{li}) = \theta_{li} g_t \tag{2}$$

Appendix A microfounds this multiplicative functional form with a model of downstream retail in which consumers have CES utility over varieties and retailers engage in Cournot competition. In this leasing model, θ_{li} corresponds to a function of retailer costs and consumer preferences across varieties in the CES demand model. θ_{li} is increasing in consumer preferences for a tenant's own product, and decreasing in the tenant's marginal costs. Variation in our model's aggregate state g corresponds to pure demand shocks in the CES retail demand model.[22]

Existing tenant continuation/exit payoffs Incumbent tenant i observes the current period's aggregate state g_t and an idiosyncratic opportunity cost ϕ_{it}. The opportunity cost reflects the payoff of engaging in any other possible activity; for example, it might represent the profits the tenant

the transition of the aggregate state in terms of μ_g since μ_g is the long-run mean of g_t.

[21]Since we assume tenants last for only 1 lease, θ_{il} is not a state variable for the tenant. However, since landlords have multiple tenants over time, they treat θ_{li} as a state variable which evolves deterministically over the course of a lease and only is uncertain when they are searching for a new tenant.

[22]We do not separately estimate landlord marginal costs and consumer preference parameters because we lack data on retail sales or profits.

could earn from starting a different retail business in another space. Given g_t and ϕ_{it}, they choose whether to continue or exit. If they stay in, they earn net profits (gross profits minus rent), continue to the next period, and repeat the process again. We denote tenant i's value function when entering state s_{it} by $W(s_{it};\theta_{li})$.

The conditional value of choosing to continue on lease is

$$W^{continue}(j_{it}, r_{it}, g_t; \theta_{li}) = \pi(g_t;\theta_{li}) - r_{it} + \beta\mathbb{E}_t\big[W(j_{it}+1, r_{it}, g_{t+1};\theta_{li}) \mid g_t\big] \tag{3}$$

If the tenant chooses to exit, they immediately cease operations, pay rent r_{it}, and exit, receiving their opportunity cost ϕ_{it} in lieu of a continuation value. The conditional value of exiting is therefore:

$$W^{exit}(j_{it}, r_{it}, g_t, \phi_{it}; \theta_{li}) = -r_{it} + \phi_{it} \tag{4}$$

In the final period of the lease, we assume the tenant must exit. They simply operate and receive flow profits, so their terminal value is

$$W(T, r_{it}, g_t, \phi_{it}; \theta_{li}) = \pi(g_t;\theta_{li}) - r_{it} \tag{5}$$

For all ages $j < T$ (prior to the final period of the lease), the tenant's value $W(s_{it}, \phi_{it}; \theta_{li})$ encodes the value of choosing between staying in and exiting:

$$W(s_{it}, \phi_{it}; \theta_{li}) = \max\{W^{continue}(s_{it};\theta_{li}), W^{exit}(s_{it}, \phi_{it};\theta_{li})\} \tag{6}$$

The tenant's continuation value of staying operational is given by the expected value of making the same choice at period $t+1$, where the expectation is taken over the $t+1$ draws of the aggregate state and opportunity costs:

$$\mathbb{E}_t[W(s_{i,t+1}, \phi_{i,t+1};\theta_{li}) \mid g_t] = \mathbb{E}_{g_{t+1},\phi_{it+1}}\big[\max\{W^{continue}(s_{i,t+1};\theta_{li}), W^{exit}(s_{i,t+1}, \phi_{i,t+1};\theta_{li})\} \mid g_t\big] \tag{7}$$

Existing tenant's continuation/exit probabilities In state s_{it}, tenant i will choose to exit if and only if their current opportunity cost draw exceeds a threshold $\phi^*(s_{it};\theta)$:

$$W^{exit}(j_{it}, r_{it}, g_t, \phi_{it}; \theta_{li}) > W^{continue}(j_{it}, r_{it}, g_t; \theta_{li}) \implies \phi_{it} > \pi(g_t, \theta_{li}) + \beta \mathbb{E}_t[W(s_{i,t+1}) \mid s_{it}] \equiv \phi^*(s_{it}; \theta_{li}) \tag{8}$$

so his exit probability is

$$p^x(s_{it}; \theta_{li}) = 1 - F_j^{\phi}(\phi^*(s_{it}; \theta_{li})) \tag{9}$$

where F_j^{ϕ} is the cdf of the opportunity cost distribution for tenants on leases of age j. Intuitively, the opportunity cost represents the value of an activity that the tenant could be doing if they were not running their current retail business (for example, they could start and operate a different retail business in New York City). If the tenant were to close their current business earlier (with a lower j and thus more time left on their current lease), they would be able to start on their other business sooner, and so the opportunity cost of the current business is higher. With only one period left to go on the current lease, there is not much to gain from exiting the current lease early to start the new business. Therefore, the opportunity cost distribution shifts down as the lease ages. Indeed, allowing the opportunity cost distribution to depend on the lease age allows us to match the empirical distribution of tenant ages at exit, which we observe in Figure 7.[23]

For estimation, we assume that when a lease is of age j, the tenant draws their opportunity cost from an Exponential distribution with a mean equal to

$$\sigma_{\phi}(T - j) \tag{10}$$

[23] In our model, tenants' continuation value functions are decreasing in their lease age: the older the lease is, the fewer future periods the tenant has remaining in which to earn profits. If we assumed opportunity costs were drawn from the same distribution every period regardless of age, tenants would only exit at the end of their lease. This is not what we observe in the data: Figure 7 shows that most tenants exit fairly early on in their lease. The empirical distribution of tenants' age at exit is consistent with Jovanovic (1982)'s theory of industry dynamics in which tenants learn about their own quality over time. In such a learning model, the high-quality tenants survive while the low-quality tenants fail. A full learning model is difficult to estimate without data on firm sales or profits (which we lack), and is also not necessary in a model primarily focused on lease pricing (in which landlords must set prices before they have the opportunity to learn from their tenant's experience). Therefore, rather than assuming landlords and tenants learn about θ_{li} over time, we assume persistent but fixed unobservable match quality heterogeneity, and allow the opportunity cost distribution to vary with tenant age.

so the mean opportunity cost is highest at the beginning of the lease, and lowest at the end. The mean opportunity cost has this downward trajectory over the course of the lease for both an intuitive and a mechanical reason. Mechanically, the mean of the opportunity cost distribution is downward-sloping with respect to the age of the lease in order to mirror the downward-sloping trajectory of the tenant's continuation value. Since tenants exit at the end of the lease period with a terminal value of zero, their value function is (in expectation) highest at the beginning of the lease (when j is low) and lowest at the end (when $j = T$).

Potential tenant's participation constraint If the tenant accepts a rent offer, they will move in at the beginning of the following period (sinking their move-in cost m) and then begin a lease in its first period. The tenant's value of accepting rent offer r in period t (for move-in in period $t+1$) is therefore given by

$$W^{accept}(r, g_t; \theta_{li}) = \beta\Big(- m + \mathbb{E}_t\big[W(1, r, g_{t+1}, \phi_{i,t+1}; \theta_{li})) \mid g_t\big]\Big) \tag{11}$$

and we normalize the tenant's value of rejecting a rent offer to 0.

We can characterize the tenant's value function and behavior using the following propositions, whose proofs can be found in Appendix B.

Proposition 4.1. *The tenant value function $W(j, r, g; \theta)$ is strictly decreasing in rent r for all j, g, θ.*

Proposition 4.2. *The exit probability $p^x(j, r, g)$ is strictly increasing in r for all $j < T, g, \theta$.*

4.3 Landlord Payoffs and Behavior

Landlord's on-lease payoffs We now turn to describing the landlord's payoffs and behavior. In the following desciption of landlord actions and payoffs, we use i to refer to a landlord's incumbent tenant, and n to refer to a new potential tenant drawn during the period.

A landlord l who is on lease has state $s_{lt} = (j_{lt}, r_{lt}, g_t, \theta_{li})$, where j_{lt} is the age of the current lease (or an indicator for vacancy), r_{lt} is the contractual rent l collects in period t, and g_t is aggregate retail demand in period t. While θ is a fixed match quality for each landlord-tenant pair, it varies across tenants which occupy the landlord's space at different times. Therefore, landlords treat θ_{li} as a state variable which evolves deterministically over the course of a lease and only is uncertain when they are searching for a new tenant.

In all periods in which the landlord has a tenant, the landlord earns contractual rent as a flow payoff. Their continuation value depends on the tenant's continue/exit decision. If the tenant continues (which occurs with probability $1 - p^x(s_{lt})$), the landlord's contination value is simply the discounted expected value of a lease one period older, with the same rent and the same tenant. If the tenant exits (either because their lease expires or the tenant endogenously exits), then the landlord searches, receiving the value of searching given the current value of aggregate retail demand ($U(g_t)$, which we define below). The landlord's value function when in state s_{lt} can thus be written:

$$V(s_{lt}) = r_{lt} + (1 - p^x(s_{lt}; \theta_{li}))\Big(\beta \mathbb{E}_t[V(s_{l,t+1}) \mid s_{lt}]\Big) + p^x(s_{lt}; \theta_{li})U(g_t) \tag{12}$$

Upon reaching the final period T of a lease with a tenant, the tenant pays the contractual rent and the landlord searches for a new tenant. Therefore the value of a lease in its final period is simply equal to the flow rent payment plus the value of searching:

$$V(T, r_{lt}, g_t, \theta_{li}) = r_{lt} + U(g_t) \tag{13}$$

Landlord's payoffs from search We model search very simply: in each period in which the landlord searches (when they enter the period vacant, j_{lt} = vacant; their incumbent tenant's lease is expiring, $j_{lt} = T$; or the incumbent tenant has announced their intention to exit this period, $\phi_{i,t} > \phi^*(s_{it})$), they encounter ("match with") a potential tenant with probability p^m. The model requires only that the probability of matching is independent of the landlord's idiosyncratic state $(j_{it}, r_{lt}, \theta_{li})$; it may depend on the aggregate state g_t. We describe the matching function we choose for our empirical specification in subsection 4.4. Conditional on matching with a potential tenant, the pair's match quality is drawn randomly from the exogenous match quality distribution F^θ. Once a searching landlord has observed a new potential tenant's move-in cost, they choose rent to maximize the value of a first-period lease with that tenant subject to the tenant's participation constraint. Therefore, the value of accepting a tenant of quality θ is:

$$
\begin{aligned}
V^{accept}(g_t, \theta) = \max_r \beta \cdot \mathbb{E}_t[V(1, r, g_{t+1}, \theta) \mid g_t] \\
\text{subject to} \quad W^{accept}(r, g_t; \theta) \geq 0
\end{aligned} \tag{14}
$$

Let $r^*(g_t, \theta)$ denote the solution to this problem.

If a searching landlord rejects a tenant, they simply repeat the search process again in the next period. Searching is costless and carries no flow payoff, so landlords who can search always do. Therefore, the conditional value of rejecting a tenant for a vacant landlord is:

$$
V^{reject}(g_t) = 0 + \beta \cdot \mathbb{E}_t[U(g_{t+1}) \mid g_t] \tag{15}
$$

where the expectation is taken over next period's aggregate state.

If the landlord matches with a tenant, they observe θ_{ln} and choose to either reject n outright or make them a rent offer. Regardless of the landlord's decision about n, they earn flow payoffs r_i according to the contract with their incumbent tenant i (if they have one) and then i exits.

The value of searching, before observing θ_{ln}, is therefore

$$
U(g_t) = p^m(g_t)\mathbb{E}_\theta\Big[\max\{V^{reject}(g_t), V^{accept}(g_t, \theta)\}\Big] + (1 - p^m(g_t))V^{reject}(g_t) \tag{16}
$$

Landlord leaseup policy The landlord accepts a tenant with match quality θ whenever $V^{accept}(g_t; \theta) \geq V^{reject}(g_t)$. The leaseup probability is therefore given by

$$
p^l(g_t) = Pr(V^{accept}(g_t; \theta) \geq V^{reject}(g_t)) \tag{17}
$$

4.4 Search and Matching

We allow the matching probability p^m to depend on downstream retail demand g_t. We will therefore refer to p^m as the "matching function" and assume it is given by the following functional form:

$$
p^m(g_t) = \frac{\exp\big(\lambda_0 + \lambda_g(g_t - \mu_t)/\sigma_g\big)}{1 + \exp\big(\lambda_0 + \lambda_g(g_t - \mu_t)/\sigma_g\big)} \tag{18}
$$

The parameter λ_0 adjusts the average probability of drawing a tenant each period. When λ_0 approaches infinity, the match probability p^m asymptotes to 1; when λ_0 approaches negative infinity, the probability of drawing a tenant each period asymptotes to 0. The parameter λ_g allows the strength of search frictions to vary across levels of the aggregate state. Setting $\lambda_g = 0$ forces search frictions to have the same strength regardless of the level of downstream retail demand. By contrast, a positive value of λ_g makes it easier for landlords to find a tenant when downstream retail demand is high (and vice versa if λ_g is negative). One interpretation of a positive value of λ_g is that more retailers enter and look for retail space when downstream demand is high. Another interpretation is that search frictions are lower when downstream demand is high. Since we do not observe searching tenants before they lease a space, we cannot distinguish between the two interpretations.

By extension of its effect on matching probabilities, λ_g also affects the degree of dispersion in landlord value functions V^{reject} and U across levels of the aggregate state. When $\lambda_g = 0$, the landlord's outside option is highest in the highest retail demand state and lowest in the lowest retail demand state. This is because tenants earn higher profits when retail demand is high, so landlords can command higher rents. Even though the aggregate state is likely to mean-revert during the term of the lease, the more that mean-reversion can be discounted, the higher the total rent that landlords can extract. When $\lambda_g > 0$, all else equal, V^{reject} increases for the highest levels of the aggregate state and decreases for the lowest levels of the aggregate state. This increases the landlord's value of waiting in the good state. When λ_g is sufficiently low, the model can generate countercyclical vacancy rates. For high enough values of λ_gs, the model can generate procyclical vacancy.

4.5 Equilibrium

Equilibrium is defined by a set of rents, lease-up probabilities, and exit probabilities such that landlords accept only tenants who are preferable to vacancy (conditional on the aggregate state), rents maximize $V^{accept}(g, \theta_{li})$; tenants exit if and only if $\phi_{it} > \phi^*(s_{it}; \theta_{li})$, and all agents have rational expectations.

4.6 Solving the Model

Because our model is a combination of two single-agent dynamic optimization problems (one each for landlords and tenants), we are able to solve the tenant's and landlord's problems sequentially. This improves the speed of computation dramatically. The key assumption necessary to make this simplification is is that once rent has been set in the contract, the tenant's exit policy does not depend on the landlord at all. This assumption would be violated if, for example, rents could be re-negotiated after leases were signed. However, anecdotal evidence suggests that rents are almost never renegotiated during the term of a lease in New York City retail leasing markets[24]

We first discretize the state space. We discretize the aggregate state g into 10 bins, and allow for 40 rent values and 40 match quality values. We compute discrete approximations to the match quality distribution given the parameters, and compute the aggregate state transition matrix. We then solve each tenant's value and policy functions for each state $s = (j, r, g)$ and each match quality θ by backward induction from the end of the lease. Finally, we perform a contraction mapping to solve the landlord's value function, taking tenant behavior as given. We start the contraction mapping with a guess of the value of searching, $U(g)$, for each value of g. Given that guess, we backward-induct the value of a lease with each match quality θ at each rent r, from the final period ($j = T$) back to its first period ($j = 1$). Then we find the rent which maximizes the landlord's value of a lease in its first period for each match quality, subject to that tenant's participation constraint. Finally, we update the guess of $U(g)$ and repeat until convergence.

4.7 Identification

A central contribution of the model is that it allows us to separately identify search frictions (the parameter λ_0 of the matching function) from the option value landlords derive from waiting for better-quality tenants (the variance of the match quality distribution, σ_θ). If a searching landlord fails to sign a lease in any given period, we are unable to distinguish in our data whether this failure was due to (1) the landlord facing search frictions and simply not encountering potential tenant, or (2) the landlord drawing a potential tenant but finding their match quality unacceptably low. Both mechanisms lead to low observed lease-up rates in the data. However, they have very different implications for lease pricing and the observed distribution of rents.

To illustrate this, compare the observed rent distributions in two scenarios with identical match quality distributions but different search frictions. Figure 2 illustrates this comparison. The figure

[24]The Covid-19 pandemic was a rare exception to this norm: retail foot traffic collapsed so dramatically across the city that landlords often were willing to reduce rents for short periods of time. However, the pandemic was unanticipated and occurred after our study period, so we assume rents cannot be renegotiated during the term of the lease.

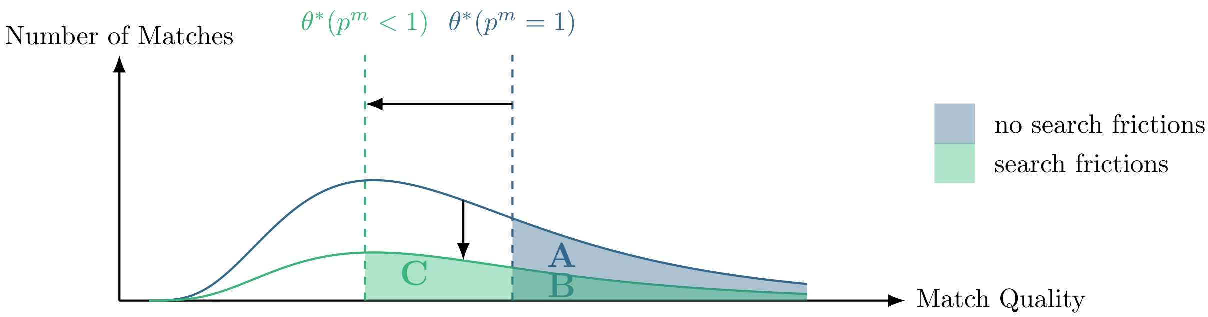
depicts match quality on the x axis and the number of matches predicted by the model on the y axis. The upper (dark blue) curve is the probability density function of the match quality distribution when the probability of encountering a tenant every period, p^m is equal to 1 (that is, there are no search frictions). In this case, all landlords draw a tenant every period, so the number of matches at each match quality is simply proportional to the density of the match quality distribution. The lower (green) curve depicts the number of matches when search frictions are present (i.e. the probability of encountering a tenant every period is less than 1). In this case, fewer landlords draw tenants, so the total number of matches is lower at every match quality. However, the shape of the match quality distribution remains unchanged. The two vertial lines illustrate the landlord's accept/reject threshold in each case. When search frictions are present, landlords know that they may not draw a tenant every period, so they need to accept lower-quality tenants to achieve the same probability of leasing up. As a result, as search frictions increase, the accept/reject threshold shifts leftward.

In the scenario without search frictions, the landlord accepts all tenants with match quality above the right (dark blue) vertical line. The area under the match quality PDF to the right of that line therefore gives the number of leases signed. This corresponds to the areas labeled A and B. In the scenario with search frictions, the landlord accepts all tenants with match quality above the left (green) vertical line. The area under the match quality PDF to the right of that line therefore gives the number of lease-ups in the presence of search frictions. This is given by the areas labeled B and C. The illustration makes clear that the distribution of rents changes: in the presence of search frictions, landlords accept lower-quality tenants, so the distribution of observed contractual rents shifts leftward. This co-variation in the rent distribution and the number of observed matches separately identifies search frictions separately from the match quality distribution. The assumption that the match quality distribution is lognormal aids identification by reducing the number of parameters that need to be estimated (relative to a nonparametric match quality distribution).

The variance parameter of the match quality distribution is identified from landlords' lease-up probabilities conditional on matching. In particular, the probability of lease-up conditional on matching must equal the area underneath the match quality PDF to the right of the landlord's threshold.

In estimation, we also recover the mean of the match quality distribution μ_θ, the variation of search frictions over the business cycle (λ_g), the parameters associated with the transition of the

Figure 2: Identification of search frictions and match quality distribution



aggregate state $(\rho_g, \mu_g, \sigma_g)$, and the opportunity cost distribution (σ_ϕ). As discussed in section 4.4, the parameter λ_g allows the strength of search frictions to vary across levels of the aggregate state. It is therefore identified by variation in the lease-up probability across levels of the aggregate state. The parameters associated with the aggregate state are identified from its own serial correlation. The opportunity cost parameter σ_ϕ is identified from exit rates.

The move-in cost parameter m is formally identified by the average ex-post duration of a landlord-tenant relationship. Recall that landlords set rent by making tenants indifferent between accepting and rejecting their rent offer. In other words, the tenant's expected discounted profits over the amount of time they expect to stay in the market must equal their move-in cost. Under the assumption of rational expectations, then, the expected amount of time the tenant expects to earn profits identifies the move-in cost.

5 Estimation

We estimate a discrete approximation to the continuous model described in section 4. Estimation proceeds in 2 stages. In the first stage, we estimate the aggregate state process by maximum likelihood. As we saw in the model section, this process is the primary driver of changes in tenant profits, and thus landlords' leasing policies, over time. In the second stage, conditional on the estimated aggregate state process, we recover the remaining model parameters: the distribution of unobserved tenant heterogeneity (μ_θ and σ_θ), the parameter of the scrap distribution σ_ϕ, and the matching function parameters λ_0 and λ_g.

5.1 Estimating the aggregate state transition process

The aggregate state variable in the model, g_t, represents exogenous downstream retail demand. Our retail tenants fall into three main categories: retail trade, consumer services, and business services. The retail trade category includes to NAICS codes beginning with 44 and 45, including for example furniture, hardware, apparel, grocery and hobby stores. Consumer services includes tenants who provide dining, hospitality, entertainment, repair, or personal services. It includes NAICS codes beginning with 7 and 8, such as restaurants, coffee shops, bars, salons, spas, dry cleaners, and shoe repair stores. Business services include bank branches, real estate agencies, doctors' offices, and lawyers' offices. This final group of tenants fall in NAICS codes beginning with 5 and 6.

We use industry-level GDP from the Bureau of Economic Analysis (BEA) as a measure of aggregate retail demand. The GDP measure is quarterly and available at the national level from 2005 to 2019. It covers all retail categories, including traditional goods retailers (NAICS codes 44-45), restaurants and bars (NAICS code 722), personal service providers (NAICS code 812), and business services (including bank branches, real estate agencies, and doctors' offices). Unlike retail sales series from the Census Bureau, the BEA GDP data also include personal and business services (which are present in our Live XYZ and CompStak datasets). Furthermore, industry GDP measures value added rather than sales, which is more relevant for retailers' profits and thus their willingness to pay for retail space after other expenses.

Furthermore, we rely on this national data series to avoid endogeneity concerns. Local demand indicators (e.g., NYC retail sales or neighborhood-level pedestrian footfall) are potentially endogenous to retail vacancy rates. BEA industry GDP, while not immune to endogeneity concerns, reflects national rather than local shocks and thus mitigates this risk. We assume that national retail demand shocks are correlated with local retail demand shocks, but not perfectly so.[25]

Before taking the average of GDP across industries, we adjust the BEA's industry-level GDP measure for the goods industry in order to account for the growth in e-commerce over our sample period. We compute the national, quarterly, brick-and-mortar share from the Monthly Retail Sales report, and then apply it to each quarter's goods GDP measure.

[25] We explored using other data series, including foot traffic, retail sales, and subway ridership data. However, these data series are either not available at a high frequency over our sample period, are potentially endogenous to local retail vacancy rates, or do not correlate with business cycle fluctuations we think are important to capture.

Our aggregate state variable requires a few modifications before it can be brought to our structural model. Our model is stationary, so we first remove a linear time trend from our aggregate state measure. We then convert the level of real GDP to per-square-foot units using a two-step process. First, we use establishment counts by NAICS code from the County Business Patterns to obtain output per establishment. Then, we use the average square footage of a storefront in each category in our CompStak sample to compute average output per square foot.

Using national data for our aggregate state variable constrains our estimation by ruling out variation in aggregate uncertainty across neighborhoods. For example, retailers in the Financial District and on the Upper West Side face the same g at all times and have the same expectations about the future evolution of g. As described in section 2, we feel that using national retail demand helps ensure that our aggregate state variable is not itself endogenous to local vacancy rates. Furthermore, we feel the assumption of a common aggregate state across neighborhoods is not terribly restrictive for two reasons. First, the multiplicative structure of tenant gross profits. The mean and variance of quality θ varies freely across neighborhoods, so tenants in different markets face systematically different flow profit distributions. Similarly, the unobservable type distribution absorbs level differences between our aggregate state measure (GDP per square foot) and tenant profits ($\pi(g, \theta)$ in the model). Second, consumers are mobile within New York: a shopper who lives in the Financial District may travel to the Upper West Side in the course of their day and make retail purchases there, and vice versa. This mobility means that local retail demand is likely to be highly correlated across neighborhoods within New York City.

We estimate the parameters of the AR(1) process governing the aggregate state using maximum likelihood. The the empirical series is shown in figure 8 and estimated AR(1) parameters are reported in table 3. The aggregate state is highly persistent, with an estimated persistence parameter of 0.95. The average value of the aggregate state is \$333.75 per square foot. Our sample period contains a substantial business cycle corresponding to the Great Recession. The aggregate state only recovers to its mean value in about 2015, just 2 years before the Live XYZ dataset begins.

5.2 Estimating remaining model parameters

We estimate the remaining parameters separately for each market by matching the simulated method of moments. Specifically, for each market, we recover the parameters governing the distribution of unobserved tenant heterogeneity (μ_θ and σ_θ), the scrap value distribution (σ_ϕ), and the matching function (λ_0 and λ_g).

We do not observe each agent's complete state vector (match quality is by definition never observed, and we only observe the age of a tenant and the rent they pay if we happen to observe their lease in the CompStak dataset, which represents a selected sample of all storefronts in Manhattan). This creates two challenges for estimation. First, we cannot compute empirical choice probabilities for landlords or tenants in every state. As a result, we cannot use traditional conditional choice probability (CCP) methods to estimate our model as Brancaccio et al. (2020a) or Vreugdenhil (2020) do. Second, we do not observe the initial distribution of individual landlord states at the beginning of our sample period. This creates an initial conditions problem. We discuss each of these challenges in turn below.

Instead of using CCP methods, we estimate the parameters using a full solution method, matching four aggregate moments over up to 60 quarters each.

Rent moments We match the average and standard deviation of contractual rents for leases signed in each quarter. We construct this moment from the sample of leases in our leasing dataset, which contains leases signed between 2005Q1 and 2019Q4. We are therefore able to match average contractual rents over this entire 60-quarter period.

Exit rates We also target the path of aggregate exit rates (the share of incumbent tenants who exit each quarter) as constructed from our occupancy panel. The universal coverage and high-frequency nature of this dataset allows us to compute accurate aggregate exit rates at the neighborhood level. We match the share of all incumbent tenants who exit in each quarter rather than matching the probability of exit conditional on age to avoid concerns about the selection of our matched dataset.

While individual landlords' exit policies depend only on their current state (and expectations over future states conditional on the current state), the aggregate exit rate depends on the composition of agents in the market. The overall market exit rate (the share of incumbent tenants who exit in a given period) is obtained by integrating $p^x(s_t)$ over the distribution of tenants in the market, conditional on the aggregate state g:

$$\mathbf{p^x}_t = \int_{j,r,\theta} p^x(j,r,g;\theta)dF_t(j,r,\theta) \tag{19}$$

where $F_t(j,r,\theta)$ is the joint distribution of lease age, contractual rents, and match quality at time t. This joint distribution depends on the last T periods worth of history: there may still be tenants in the market who signed leases T periods ago, and the share of leases of each age j which survived until period t depends on the entire path of the aggregate state from period $t-j$ until t.

Since tenant exit rates depend on the age of the lease, the share of tenants exiting at time t depends on the distribution of lease ages in the economy. For example, number of leases of age 2 at time t depends on the number of vacancies at time $t-2$, which itself depends on exit rates and lease-up rates at time $t-3$, and so on. This means that given our chosen state space, our model generates a non-Markov path of aggregate exit rates.

Throughout the paper, we use the term "probability" to refer to an individual agent and "rate" to refer to an aggregate quantity. So, for example, we use "exit probability" to refer to an individual tenant's probability of exit in a given state. We use the term "exit rate" to refer to the share of incumbent tenants who exit in a given period. In mathematical notation, we use bold text to indicate aggregate quantities. For example, individual tenant exit probabilities ($p^x(s_t)$) are a function of the current state s_t only.

Lease-up rates We also target the path of aggregate lease-up rates. We define the lease-up rate as the share of searching landlords who sign a lease with a new tenant in a given quarter. We assume that a storefront has leased up in period $t-1$ if a new tenant appears in period t, and assume that they are searching in t if they are either vacant or their incumbent tenant exits during period t.

The lease-up rate is given by the integral of the individual landlord lease-up probabilities over the distribution of searching landlords in the market:

$$\mathbf{p^l}_t = \int_{j,r,\theta} p^l(j,r,g;\theta)dF_t(j,r,\theta) \tag{20}$$

Vacancy rates Our next moment is the vacancy rate in each quarter, which is simply the number of vacant storefronts in the market divided by the total number of storefronts. We concatenate the time series of vacancy rates reported by the Comptroller's office with the time series of vacancy rates that we calculate from the Live XYZ dataset. The property tax filings ask landlords to report on vacancy rates as of January 5, so we assume the reported vacancy rates correspond to first-quarter vacancy rates, and linearly interpolate to estimate vacancy rates in the intermediate quarters for 2007-2017. There are 2 community districts for which we do not observe vacancy rates for any contained zip codes. For these zip codes, we substitute Manhattan's overall vacancy rate.

It is important to note that the vacancy rate is a non-Markov object. The vacancy rate at the beginning of period t depends on the vacancy rate in period $t-1$, the share of vacant landlords who signed leases in $t-1$, and the share of occupied landlords in period $t-1$ whose tenants exited and who failed to sign new leases. Specifically, the transition of the vacancy rate is given by

$$\nu_t = \nu_{t-1} - \mathbf{p}^{\mathbf{l}}(g_{t-1})\nu_{t-1} + (1-\nu_{t-1})(1-\mathbf{p}^{\mathbf{l}}(g_t))\mathbf{p^x}_t \tag{21}$$

That our model generates non-Markov aggregate vacancy, lease-up, and exit rates is a choice: we could expand the state space to include the T most recent values of the aggregate state. However, this dramatic expansion of the the size of the state space would make our model much more difficult to solve. Successfully estimating a model with a larger state space would also require us to observe more storefronts over a longer period of time, so that we could match lease-up and exit probabilities conditional on age and contractual rent. We therefore choose to keep our state space small, and instead match non-Markov aggregate moments.

Correlational moments Finally, we match the correlations of the aggregate state with each of the moments described above: the exit rate, leaseup rate, vacancy rate, and average rents.

Simulated method of moments estimation Our moment condition corresponding to period t simply takes the difference between the model-predicted moments and the observed moments. To construct the SMM moment condition, we stack the period-level moment conditions into a vector, and then append the correlation moments. Our estimated parameters are those that minimize the weighted mean squared error of the model-predicted moments. We weight each moment by the reciprocal of the number of quarters for which it is observed. For example, we observe 60 quarters of average rent for each market, so the average rent moments each receive a weight of $1/60$. Since we are estimating only 5 parameters for each market, the model is over-identified.

Calibrated parameters In our model, the move-in cost m captures not just the fixed cost of renovating a space, but also any up-front investment involved in starting their store. We lack data on move-in costs, but market participants indicate that renovation costs are often between \$300 and \$400 per square foot. To account for additional costs of starting a retail business, including advertising, hiring costs, and obtaining permits, we calibrate move-in costs at \$650 per square foot.

Initial Condition The second challenge we address in our estimation is a variant of the initial conditions problem of Heckman (1981). In order to match aggregate moments, we need to start simulating our model at some initial distribution of individual landlord states. This state vector includes the aggregate state g, as well as (for occupied storefronts) the age of the lease j, the contractual rent r, and the tenant's type θ. Because we do not observe the distribution of individual states at the beginning of our sample period, we use our model to simulate it. Our approach is similar to those taken by Pakes (1986) and Ho and Lee (2022), but adapted to account for aggregate uncertainty.

While we do not observe landlords' full individual state vectors, we do observe the aggregate state g beginning in 2005Q1. We therefore construct an algorithm to draw paths of landlord states which are consistent with the observed aggregate state from 2005Q1-2019Q4.

First, given a draw of the parameters, we solve the model and simulate it for many periods, starting all landlords as vacant. The purpose of this simulation is to reach and explore the recurrent class of landlord states (which Ericson and Pakes (1995) show exists). This simulation allows us to compute the long-run distribution of individual landlord states. Next, we use the estimated aggregate state parameters (ρ_g, σ_g, and μ_g) to simulate a large number of pre-period aggregate state paths that all end at the the observed aggregate value in 2005Q1. For each simulated aggregate state path, we assume there is a fixed number of landlords, and draw their initial states from the long-run distribution of individual landlord states, conditional on the aggregate state. From this initial state, we simulate the model forward along each aggregate state path, transitioning in 2005Q1 from following the simulated path to the observed path of the aggregate state. In each period between 2005Q1 and 2019Q4, we compute the lease-up rate, exit rate, average rent on new leases, and vacancy rate.

5.3 Results

Figure 9 illustrates the estimated match quality distributions for each neighborhood as box-and-whisker plots. The figure shows that on average, the neighborhoods where landlord-tenant pairs are able to generate the most revenue are the Upper East Side (which contains Madison Avenue), Midtown (containing Times Square and 5th Avenue), and the Lower West Side (which contains SoHo and the West Village). These are the high-rent "destination" retail neighborhoods which generally serve tourists in addition to locals. Madison Avenue, 5th Avenue, and SoHo are primarily commercial districts and play host to many luxury retailers' flagship stores. By contrast, the Upper West Side, Midtown West, Midtown East, and Lower East Side are more residential, so their retail mix is composed of more restaurants and personal services establishments. The Financial District, the neighborhood where we estimate the lowest match quality between landlords and tenants, historically was dominated by office buildings, but in recent districts has become much more residential. As such, its retail mix also caters more to locals' needs.

The remaining parameter estimates for each market are reported in table 4. Table 4 shows that in most markets, the unconditional probability that searching landlords get to draw a tenant is 94% or higher. This is consistent with the fact that New York City is a market with high demand for retail space (as indicated by the fact that rents are high). The high match probability indicates that search frictions are not so strong that landlords are unable to find potential tenants to sign leases with. This does not mean that search frictions are negligible, however – in a truly frictionless world, landlords would be able to choose the best match from a very large set of potential tenants every period. If landlords could inspect an infinite number of potential tenants each quarter, they would not need to wait for a tenant with high θ to arrive – one would always be available to sign a lease immediately. Rather, our estimates suggest that landlords are able to inspect a potential tenant every quarter.

The Upper East Side is the only neighborhood in which our estimates suggest that landlords may not encounter a potential tenant every quarter – in fact, we estimate that searching landlords on the Upper East Side only match with a potential tenant with 32% probability each quarter. This occurs because vacancies on the Upper East Side occur less frequently than in the rest of the city, but last longer on average. However, the model fit in general for the Upper East Side is poor, because its average rents are so much higher than any other market in our data. We believe this is due to the fact that these tenants are not small businesses serving local residents, but rather are large (usually luxury) chain tenants who view their flagship Madison Avenue stores as a status symbol that they are willing to operate at a loss. Though we lack data on renewal rates, many Madison Avenue stores have occupied their spaces for a long time and renewed their leases multiple times.[26]

6 Quantifying the Sources of Vacancy

In this section, we perform a series of counterfactual exercises with our structural model to quantify the degree to which aggregate uncertainty in downstream demand and match quality each contribute to long-run vacancy rates. In each exercise, we vary the parameters associated with each source of landlord option value to reduce the intensity of its effects. We find that, while reducing the variance of downstream retail demand shocks by 50% has a negligible effect on the long-run vacancy rate, reducing the variance of match quality by 50% reduces long-run vacancy rates by about 33%.

First, we reduce the role of aggregate demand uncertainty by scaling down σ_g from its estimated quantity to 0, while holding μ_g (the long-run average of downstream retail demand) constant. Simulating the model under the assumption that the aggregate downstream retail demand state stays at its mean value forever is a helpful benchmark for two reasons. First, removing aggregate uncertainty completely shuts down variation in each tenant's individual profits over time. This means that the only uncertainty tenants face each period comes from their idiosyncratic opportunity cost draws. Furthermore, there is no longer any dispersion in landlords' value of rejecting a potential tenant across periods. This means that, when $\sigma_g = 0$, landlords have the same match quality threshold in every period.

Second, we reduce the role of uncertain future match quality by scaling down the variance of the match quality distribution from its estimated level, while holding the mean match quality value constant. Intuitively, when $\sigma_\theta = 0$, there is no reason to wait for a better tenant to come along because all tenants are ex ante the same. However, vacant spaces may remain unfilled, either

[26] Our model does not explicitly account for lease renewals, since most tenants do not stay for the duration of their first lease. Relaxing the model to allow for lease renewals would lead to a higher-quality distribution of tenants in the long run, since high-quality tenants would be more likely to renew. This would likely lead to lower exit rates and higher average rents in the long run, and would probably match the Upper East Side data better.

because the landlord fails to match with a tenant or because they want to wait for downstream retail demand to improve.

Results from these exercises, as well as the case where we scale down both estimated variances at the same time, are presented for each neighborhood in figure 10. In each subfigure, the x-axis ranges from 0 to 1, and indicates the constant by which the estimated variance is multiplied in each simulation. When $x = 0$, we show the long-run vacancy rate for the case when the indicated variance is taken all the way to 0. When $x = 1$, the long-run vacancy rate is plotted from simulating the model at the estimated parameters. Note that this long-run vacancy rate does not equal the average observed vacancy rate over our study period, but rather represents the average share of properties that are vacant over a much longer horizon (thousands of periods).

Reducing the variance of demand shocks has only minor effects on the long-run vacancy rate, but reducing the variance of match quality by approximately 50% reduces the long-run vacancy rate by 33% on average. Figure 10 shows that these results hold across most neighborhoods. In most neighborhoods, completely eliminating any variation in match quality takes long-run vacancy rates to approximately zero. This is due to the fact that our estimated search frictions are small – as noted in section 5 our landlords' estimated match probabilities are almost 1 in every period – and that landlords are fairly unresponsive to the variance in aggregate demand shocks. The only neighborhood where the long-run vacancy rate is positive is the Upper East Side, where the estimated probability of matching is very low.

7 Vacancy Tax Counterfactual

Policymakers, journalists, and residents argue that storefront vacancy imposes negative externalities on pedestrians and neighborhood residents. From this perspective, vacancy taxes would seem to act like any other Pigouvian tax – they would force landlords to internalize the external costs of vacancy, leading to more efficient outcomes. However, the dynamic and frictional nature of the leasing market complicates this narrative. In this section, we use our estimated structural model to quantify the effects of a commercial retail vacancy tax on long-run vacancy rates, average rents, and welfare.

New York State Senate Bill S2005 (Jackson, 2021) proposes to tax vacant commercial storefronts in New York City an amount equal to one percent of the assessed value of the property including the vacant storefront.[27] We find that the proposed tax would increase the pace of tenant churn, but only affect the vacancy rate and average rents by a small amount. By encouraging landlords to lease up more quickly, the tax would distort the retail mix towards tenants with lower profitability, who are more likely to exit their leases early than the tenants landlords wait for in the absence of the tax. We lack data to quantify the size of externalities associated with commercial retail vacancy, but we can use our model to quantify how large such externalities would have to be in order to justify the proposed vacancy tax, under some assumptions.

Externalities associated with vacancy can come in many forms. Many pedestrians perceive of vacant storefronts as eyesores which negatively impact their experience of a neighborhood and reduce residential property values. Some are concerned that higher retail vacancy poses a threat to neighborhood safety via a reduction in "eyes on the street" (Jacobs, 1961), though there is mixed empirical evidence on whether real estate vacancy is actually associated with increased crime.[28]

Because of the market features we model, the presence of vacant storefronts is not necessarily evidence of inefficiency in the leasing market. Search frictions alone could result in a positive vacancy rate even if all tenants were the same and move-in costs were zero. To the extent that tenant profits reflect the value they generate for consumers, the social planner would want to strategically hold some storefronts vacant to ensure that high-quality tenants can find a space when they enter. In fact, it may even be socially optimal for landlords to exercise their option value in the interest of signing long-term tenants who will best meet local retail demand. Landlords' exercise of option value is inefficient only if vacancies impose an externality on other market participants (tenants, landlords, or consumers).

New York City is not the only city which has proposed a vacancy tax on some types of real estate in recent years, though the structure of the tax varies widely across municipalities. Washington, D.C. has had a vacancy tax of $5 per $100 of assessed value on vacant commercial and residential properties since 2011. In March 2020, a supermajority of San Francisco voters approved a tax on

[27] This bill was originally introduced in the 2019-2020 legislative session, but was tabled for several years during the COVID-19 pandemic.

[28] Chang and Jacobson (2017) find that a short-term mass closing of medical marijuana dispensaries in Los Angeles lead to an immediate increase in crime near those stores. They find similar results for temporary restaurant closures due to health code violations. However, in a study of urban vibrancy and crime in Philadelphia, Humphrey et al. (2020) find that neighborhoods with more vacant land have higher crime rates, but that crimes tend not to occur at vacant properties themselves.

commercial storefronts that remain vacant for more than 182 days. The tax went into effect on January 1, 2022, and is calculated based on a building's street frontage and how long the property has been vacant.[29] Oakland, California levies a fixed tax of $3000 per vacant property containing a ground-floor commercial retail vacancy.[30]

7.1 Incorporating the vacancy tax in the model

We incorporate the vacancy tax into our model by adding a flow cost τ of rejecting a tenant to equation 15 when the landlord is vacant:

$$V^{reject}(s_t, g_t) = -\tau \times \mathbf{1}(\text{vacant}_t) + \beta \cdot \mathbb{E}_t[U(g_{t+1}) \mid g_t] \tag{22}$$

Intuitively, a positive value of τ reduces the landlord's outside option, so the tax makes landlords less selective and reduces average rents. Specifically, the tax decreases the landlord's accept/reject threshold $\theta^*(g_t)$ for each value of the aggregate state and increases the probability that a searching tenant signs a lease. Since the marginal tenants have lower match quality, their participation constraint binds at lower rents. Rents offered to inframarginal tenants are unchanged, because rents hold all tenants to their individual participation constraints. Therefore, average contractual rents fall relative to a world with no vacancy tax. The vacancy tax decreases vacancy at the cost of increasing retail churn and crowding out some high-quality matches.

7.2 Vacancy Tax Consequences

In this section, we quantify the degree to which the proposed tax would reduce long-run vacancy rates and rents.

The parameter τ represents a constant tax in dollars per square foot, while the proposed tax is 1% of the assessed value of the property. For each market, we therefore set τ equal to 1 percent of the landlord's expected value of searching. We feel the value of searching is the appropriate quantity from our model to use proxy for assessed value, since New York City's assessed values for commercial

[29] The tax rate is $250 per foot of frontage in the first year, $500 in the second year of vacancy, and $1000 if the vacancy lasts for 3 or more years.

[30] Residential vacancy taxes are also being embraced by some cities, though usually because of concerns about housing supply and affordability. For example, Vancouver, British Columbia introduced a residential vacancy tax in 2017. In 2021, the Vancouver "Empty Homes Tax" was 3% of assessed taxable value.

properties are based primarily on the rents landlords report on their annual Real Property Income and Expense filings. When vacant, landlords are by definition not earning any rent, so we expect their property's assessed values to fall relative to periods in which they are occupied. We could instead use the average assessed value of buildings in each neighborhood, but New York City's publicly available data on assessed property values does not identify which buildings contain vacant storefronts.

We next simulate the model given the proposed tax to determine the quantitative impact of the tax on long-run vacancy rates and average rents. We first simulate the model for a many periods, in order to reach the recurrent class. For each level of the aggregate state g, we compute the average vacancy rate and average rent for each lease signed across all periods in which the aggregate state was g. Finally, we arrive at the long-run average outcomes by averaging the expected outcomes conditional on g over the long-run stationary distribution of g.

We show the effects of the vacancy tax on long-run vacancy rates and average rents in table 5. Long-run vacancy rates fall by 2.89 percent on average. Average rents fall by 0.44 percent. This is driven entirely by the change in landlords' acceptance threshold: the marginally accepted tenants are lower quality and so can only afford lower rents, but higher quality tenants are still held to their outside options and thus pay the same rent regardless of whether the vacancy tax is in effect or not.

The tax also increases the pace of retail churn and leads to crowd-out of high quality matches in favor of lower-quality matches which materialize sooner. Panel (a) of 11 traces out the response of both lease-up rates and exit rates to increases in the vacancy tax. The calibrated vacancy tax we use to compute the results in table 5 is shown by the vertical dashed line. We can see that, as the tax increases, both lease-up rates and exit rates increase.

In Panel (b), we show the change in the long-run share of leases associated with each value of match quality under the calibrated vacancy tax, as compared with our baseline scenario with no vacancy tax. There is no change in the long-run presence of matches with quality lower than 0.225, because matches with such low qualities are always rejected by landlords in both the baseline scenario and under the vacancy tax. However, there is a substantial increase in the share of leases with match qualities between 0.25 and 0.3. This occurs because landlords accept these matches under the tax, but did not accept these tenants if there is no penalty associated with vacancy.

7.3 Welfare

We can also use our structural model to infer the size of the externality implied by the proposed vacancy tax. To do this, we must first define a welfare function. The surplus associated with a lease in our model is given by the discounted ex post gross profits generated by the tenant, plus the scrap value they receive upon exit, less move-in costs. Mathematically, this can be written

$$\mathcal{S}\left(\{g_t\},\theta_i\right)=\underbrace{\sum_{t=t_i^1}^{t_i^x}\beta^t\pi\left(g_t,\theta_i\right)}_{\text{profits}}+\underbrace{\beta^{t_i^x}\mathbb{E}[\phi\mid\ \text{exit at age } t_i^x-t_i^1]}_{\text{scrap values}}-\underbrace{\beta^{t_i^1}m}_{\text{move in costs}} \tag{23}$$

where t_i^1 denotes the period that tenant i enters the market and t_i^x denotes the period that i exits.

We assume a social welfare function which is the discounted expected value of the surplus from all future leases (starting from some initial date $t=0$), less an externality e of vacancy each period, expressed on a per-square-foot basis. This welfare function is given by

$$\mathcal{W}(e;\tau)=\Big(\sum_i S(g_t,\theta_i;\tau)\Big)-\Big(\sum_{t=0}^{\infty}\beta^t L\nu_t(\tau)\times e\Big) \tag{24}$$

Here we have specified a social welfare function where the total externality from vacancy is linear in the number of vacant square feet ($L\nu_t$), for simplicity of interpretation. However, there is an argument to be made that the total welfare loss from vacancy is convex in vacant square footage.

7.3.1 Optimal policy with no vacancy externalities

Given the social welfare function defined in equation 24, two key frictions lead to an inefficient level of vacancies even when there is no externality from vacancy ($e=0$). These two frictions push in opposite directions: in some markets, the prevailing vacancy rate may be too low, while in others, it may be too high.

The first friction is that landlords do not capture tenants' option value once they have moved in, but the social planner does. Landlords do take tenant exit policies into account when setting the rent, but since they cannot renegotiate any part of the lease after the tenant has moved in, they do not capture any potential tenant upsides to remaining in place. If the realization of the aggregate state is higher than expected in any given period, all incumbent tenants (and the social planner) benefit through higher profits, but their landlords do not capture any of these gains. However, landlords partially capture downside risk, because tenants exit at higher rates when the aggregate state realizations are low. Because the social planner captures the upside risk, the social planner will want to fill vacancies more quickly than landlords, and thus (in the absence of any other frictions) the optimal vacancy rate would be lower than the equilibrium rate.

The second friction is that there is a wedge between the move-in costs internalized by landlords and by the social planner. Tenants (and the social planner) experience the full value of move-in costs at the beginning of the lease, when the tenant incurs them. Landlords, however, amortize move-in costs over the expected duration of the contractual relationship, via the rents they charge. By assumption, all tenants pay the same move-in costs, but the high-match-quality tenants are expected to last longer. As a result, in the absence of other frictions, the social planner would want landlords to be more selective than they would be in equilibrium and the optimal vacancy rate would be higher than the equilibrium rate.

These two frictions push the optimal tax rate in opposite directions. The friction which dominates depends on market-level parameters, and we map the optimal tax (or subsidy) across the city in figure 12. This map shows us that the move-in cost friction dominates in the high-rent markets (Upper East Side, Midtown, and Lower West Side), and the welfare-maximizing policy is actually to subsidize vacant landlords. This occurs because the high-quality tenants in these neighborhoods have such high earnings that, from the social planner's point of view, they are worth waiting for. On the other hand, the welfare-maximizing policy is a tax in the more residential, medium-rent neighborhoods, especially the Upper West Side. This indicates that in these neighborhoods, asymmetric internalization of tenant upside is the more quantitatively important friction distorting the vacancy rate from its efficient level.

To resolve inefficiencies from both frictions, an additional policy instrument is required. For example, policymakers could address the asymmetric internalization of option value friction by forcing landlords and tenants to write contracts with profit-sharing agreements rather than constant rents. They could address the wedge in move-in costs by enforcing cost-sharing of move-in costs.

7.3.2 Estimating the externality from vacancy

We back out the per-vacancy externality e for each market under the assumption that the proposed vacancy tax is the welfare-maximizing Pigouvian tax. Under this assumption, the actual externality per storefront e^* is given by

$$\tau^{proposed} = \underset{\tau}{\operatorname{argmax}}\, \mathcal{W}(e^*; \tau) \tag{25}$$

We compute welfare by simulating our model for many periods, for different simulated paths of the aggregate state. For each simulation, we keep track of the surplus generated by each lease, and the vacancy rate in each period. We then compute welfare for each simulation, and average over all simulations.

We estimate e^* by computing welfare for many combinations of e and τ. For each value of e, we find the value $\tau(e)$ which maximizes $\mathcal{W}(e; \cdot)$. We can then interpolate, for any proposed tax $\tau^{proposed}$, the e^* for which τ is welfare-maximizing.

Table 5 presents the effect of the proposed vacancy tax on long-run average vacancy rates and rents, as well as the implied externality associated with the tax. We find that on average, to justify a vacancy tax of 1% of assessed values, a vacant storefront would have to impose an externality of \$18.72 per square foot, or about thirty percent of observed average rents.

8 Conclusion

In this paper, we leverage novel data to begin to understand the key market forces that drive long-run retail vacancy rates and rents. We show that the asymmetry of landlords' and tenants' ability to commit to long-term leases, combined with tenant heterogeneity, up-front move-in costs, and search frictions, create option value for different market participants at different times, and that this option value fluctuates with the business cycle. We use our model to investigate the potential impacts of commercial vacancy taxes, a much-discussed urban policy. We view our work as a starting point for future work on the dynamics of commercial real estate leasing markets.

References

Ambrose, Brent, Patric Hendershott, and Malgorzata Klozek. 2002. "Pricing Upward-Only Adjusting Leases." *Journal of Real Estate Finance and Economics* 25 (1): 22–49.

Anenberg, Elliot, and Edward Kung. 2014. "Estimates of the Size and Source of Price Declines Due to Nearby Foreclosures." *American Economic Review* 104 (8): 2527–2551. 10.1257/aer.104.8.2527.

Benjamin, John D., Glenn W. Boyle, and C. F. Sirmans. 1992. "Price discrimination in shopping center leases." *Journal of Urban Economics* 32 (3): 299–317. 10.1016/0094-1190(92)90020-L.

Billings, Stephen B., Shawn Rohlin, and Tingyu Zhou. 2024. "The Death of Retail and Neighborhood Amenities." January. 10.2139/ssrn.4706738.

Brancaccio, Giulia, Myrto Kalouptsidi, and Theodore Papageorgiou. 2020a. "Appendix to Geography, Transportation, and Endogenous Trade Costs." *Econometrica* 88 (2): 657–691. 10.3982/ecta15455.

Brancaccio, Giulia, Myrto Kalouptsidi, and Theodore Papageorgiou. 2020b. "Geography, Transportation, and Endogenous Trade Costs." *Econometrica* 88 (2): 657–691. 10.3982/ecta15455.

Brooks, Leah, and Rachel Meltzer. 2025. "Retail on the Ground and on the Books: Vacancies and the (Mis)Match Between Retail Activity and Regulated Land Uses." *Journal of the American Planning Association* 91 (2): 192–206. 10.1080/01944363.2024.2373900.

Brueckner, Jan K. 1993. "Inter-store externalities and space allocation in shopping centers." *The Journal of Real Estate Finance and Economics* 7 (1): 5–16. 10.1007/BF01096932.

Buchholz, Nicholas. 2022. "Spatial Equilibrium, Search Frictions, and Dynamic Efficiency in the Taxi Industry." *Review of Economic Studies* 89 556–591. 10.1093/restud/rdab050.

Bulan, Laarni, Christopher Mayer, and C. Tsuriel Somerville. 2009. "Irreversible investment, real options, and competition: Evidence from real estate development." *Journal of Urban Economics* 65 (3): 237–251. 10.1016/J.JUE.2008.03.003.

Burayidi, Michael A, and Sanglim Yoo. 2021. "Shopping Malls: Predicting Who Lives, Who Dies, and Why?." *Journal of Real Estate Literature* 29 (1): 60–81. 10.1080/09277544.2021.1952050.

Caplin, Andrew, and John Leahy. 1998. "MIRACLE ON SIXTH AVENUE: INFORMATION EXTERNALITIES AND SEARCHÃ." *THE ECONOMIC JOURNAL*.

Capozza, Dennis R., and Robert W. Helsley. 1990. "The Stochastic City." *Journal of Urban Economics* 28 187–203. 10.4324/9781315240114-14.

Chang, Tom Y., and Mireille Jacobson. 2017. "Going to pot? The impact of dispensary closures on crime." *Journal of Urban Economics* 100 120–136. 10.1016/j.jue.2017.04.001.

Clapp, John M., Piet Eichholtz, and Thies Lindenthal. 2013. "Real option value over a housing market cycle." *Regional Science and Urban Economics* 43 (6): 862–874. 10.1016/j.regsciurbeco.2013.07.005.

Couture, Victor, Cecile Gaubert, Jessie Handbury, and Erik Hurst. 2021. "Income Growth and the Distributional Effects of Urban Spatial Sorting." 10.2139/ssrn.3435825.

Couture, Victor, and Jessie Handbury. 2020. "Urban revival in America." *Journal of Urban Economics* 119 (June): . 10.1016/j.jue.2020.103267.

Davidoff, Thomas. 2010. "What explains Manhattan's declining share of residential construction?." *Journal of Public Economics* 94 (7-8): 508–514. 10.1016/J.JPUBECO.2010.02.007.

Deng, Yongheng, Stuart A Gabriel, Kiyohiko G Nishimura, and Diehang Zheng. 2012. "Optimal Pricing Strategy in the Case of Price Dispersion: New Evidence from the Tokyo Housing Market." *Real Estate Economics* 40 (5): 234–272. 10.1111/j.1540-6229.2012.00347.x.

Dixit, Avinash, and Pindyck. 1994. *Investment Under Uncertainty*. Princeton: Princeton University Press.

Duguid, James, Bryan Kim, Lindsay Relihan, and Chris Wheat. 2024. "The Impact of Work-from-Home on Brick-and-Mortar Retail Establishments: Evidence from Card Transactions." *SSRN Electronic Journal*. 10.2139/ssrn.4466607.

Ericson, Richard, and Ariel Pakes. 1995. "Markov-perfect industry dynamics: A framework for empirical work." *Review of Economic Studies* 62 (1): 53–82. 10.2307/2297841.

Fréchette, Guillaume R, Alessandro Lizzeri, and Tobias Salz. 2019. "Frictions in a Competitive, Regulated Market: Evidence from Taxis." *American Economic Review* 109 (8): 2954–2992. 10.1257/aer.20161720.

Freybote, Julia, Lauren Simon, and Lauren Beitelspacher. 2016. "Understanding the contribution of curb appeal to retail real estate values." *Journal of Property Research* 33 (2): 147–161. 10.1080/09599916.2015.1135978.

Glaeser, Edward, Michael Scott Kincaid, and Nikhil Naik. 2018. "Computer Vision and Real Estate: Do Looks Matter and Do Incentives Determine Looks."Technical Report w25174, National Bureau of Economic Research, Cambridge, MA. 10.3386/w25174.

Glancy, David P., and J. Christina Wang. 2025. "Lease Expirations and CRE Property Performance." federal Reserve Bank of Boston Research Department Working Papers, Federal Reserve Bank of Boston. 10.29412/res.wp.2023.10, Series: Federal Reserve Bank of Boston Research Department Working Papers.

Grenadier, Steven R. 1995. "Valuing lease contracts: A real-options approach." *Journal of Financial Economics* 38 (3): 297–331. 10.1016/0304-405X(94)00820-Q.

Gupta, Arpit, Vrinda Mittal, and Stijn Van Nieuwerburgh. 2025. "Work From Home and the Office Real Estate Apocalypse." January. 10.2139/ssrn.4124698.

Han, Lu, and William C. Strange. 2015. "The Microstructure of Housing Markets: Search, Bargaining, and Brokerage." *Handbook of Regional and Urban Economics* 5 813–886. 10.1016/B978-0-444-59531-7.00013-2.

Heckman, James J. 1981. "The incidental parameters problem and the problem of initial conditions in estimating a discrete time-discrete data stochastic process." In *Structural Analysis of Discrete Data with Econometric Applications*, edited by Manski, Charles F, and Daniel McFadden 179–195, Cambridge, MA: MIT Press.

Hendel, Igal, and Alessandro Lizzeri. 2003. "The Role of Commitment in Dynamic Contracts: Evidence from Life Insurance." *Quarterly Journal of Economics* 118 (1): 299–328, https://academic.oup.com/qje/article/118/1/299/1917037.

Ho, Kate, and Robin S. Lee. 2022. "Health Insurance Menu Design for Large Employers." 10.2139/ssrn.3700696.

Hosios, Arthur J. 1990. "On The Efficiency of Matching and Related Models of Search and Unemployment." *Review of Economic Studies* 57 279–298, https://academic.oup.com/restud/article/57/2/279/1551634.

Humphrey, Colman, Shane T. Jensen, Dylan S. Small, and Rachel Thurston. 2020. "Urban vibrancy and safety in Philadelphia." *Environment and Planning B: Urban Analytics and City Science* 47 (9): 1573–1587. 10.1177/2399808319830403.

Jackson, Salazar. 2021. "An act to amend the tax law, in relation to the imposition of a commercial vacancy tax." https://www.nysenate.gov/legislation/bills/2021/S2005.

Jacobs, Jane. 1961. *The Death and Life of Great American Cities.* New York: Random House.

Jovanovic, Boyan. 1982. "Selection and the Evolution of Industry." *Econometrica* 50 (3): 649–670, https://www.jstor.org/stable/1912606.

Konishi, Hideo, and Michael T. Sandfort. 2003. "Anchor stores." *Journal of Urban Economics* 53 (3): 413–435. 10.1016/S0094-1190(03)00002-0.

Mooradian, Robert M., and Shiawee X. Yang. 2002. "Commercial real estate leasing, asymmetric information, and monopolistic competition." *Real Estate Economics* 30 (2): 293–315. 10.1111/1540-6229.00041.

Mortensen, Dale T, and Christopher A Pissarides. 1994. "Job Creation and Job Destruction in the Theory of Unemployment." *Review of Economic Studies* 61 397–415, https://academic.oup.com/restud/article/61/3/397/1589192.

NAREIT. 2019. "Estimating the Size of the Commercial Real Estate Market in the US."Technical report, https://www.reit.com/data-research/research/nareit-research/estimating-size-commercial-real-estate-market-us.

Office of the New York City Comptroller. 2019. "Retail Vacancy in New York City: Trends and Causes, 2007-2017."Technical report, https://comptroller.nyc.gov/wp-content/uploads/documents/Retail{_}Vacancy{_}in{_}NYC{_}2007-17.pdf.

Pakes, Ariel. 1986. "Patents as Options: Some Estimates of the Value of Holding European Patent Stocks." *Econometrica* 54 (4): 755. 10.2307/1912835.

Pashigian, B. Peter, and Eric D. Gould. 1998. "Internalizing Externalities: the Pricing of Space in Shopping Malls." *The Journal of Law and Economics* 41 (1): 115–142. 10.1086/467386.

Quan, Thomas W., and Kevin R. Williams. 2018. "Product variety, across-market demand heterogeneity, and the value of online retail." *RAND Journal of Economics* 49 (4): 877–913. 10.1111/1756-2171.12255.

Shoag, Daniel, and Stan Veuger. 2018. "Shops and the City: Evidence on Local Externalities and Local Government Policy from Big-Box Bankruptcies." *The Review of Economics and Statistics* 100 (3): 440–453. 10.2139/ssrn.2420179.

Stanton, Richard, and Nancy Wallace. 2009. "An Empirical Test of a Contingent Claims Lease Valuation Model." *Journal of Real Estate Research* 31 (1): 1–26. 10.1080/10835547.2009.12091238.

Su, Yichen. 2022. "The Rising Value of Time and the Origin of Urban Gentrification." *American Economic Journal: Economic Policy* 14 (1): 402–439. 10.1257/pol.20190550.

Tran, Uyen. 2021. "The Rise of Broadband and the Retail Apocalypse: Evidence from Consumer Packaged Goods." December. 10.2139/ssrn.3994862.

Vreugdenhil, Nicholas. 2020. "Booms, Busts, and Mismatch in Capital Markets: Evidence from the Offshore Oil and Gas Industry."

Yoshida, Jiro, Miki Seko, and Kazuto Sumita. 2016. "The Rent Term Premium for Cancellable Leases." *The Journal of Real Estate Finance and Economics* 52 (4): 480–511. 10.1007/s11146-015-9528-x.

Figure 3: Empirical Cumulative Distribution of Vacancy Durations

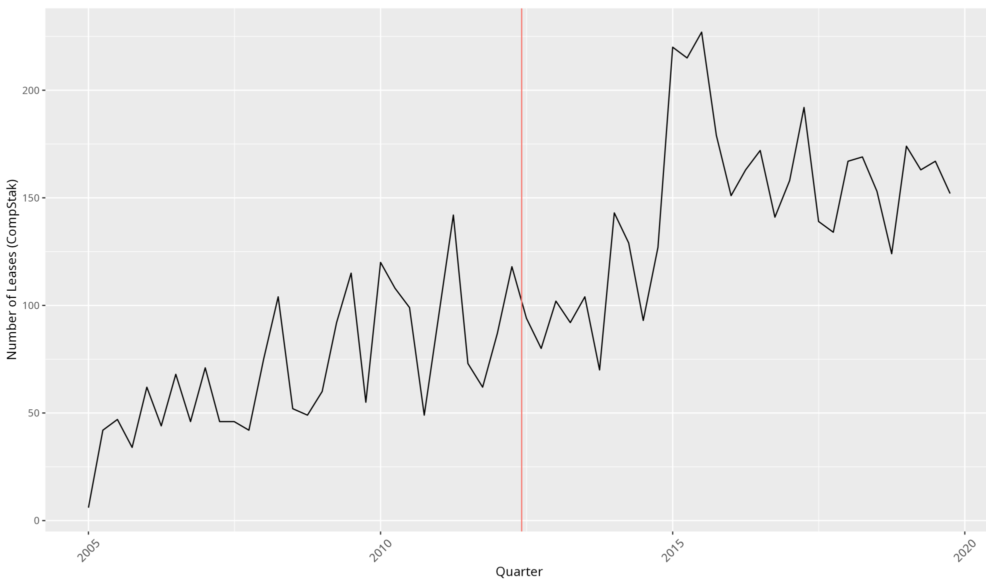
Note: Here we plot the empirical cumulative distribution of vacancy lengths in our Live XYZ dataset. This dataset is censored, and here we include all vacancies, including those storefronts which were vacant at the time they are first observed and those storefronts which were still vacant at the time they were last observed. The earliest observations in the Live XYZ dataset are from late 2015, and the last observations were in March 2020. We therefore do not observe any vacancies that last longer than four and a half years. If anything, therefore, we under-estimate the duration of the longest-lived storefront vacancies.

Table 1: Summary Statistics by Community District

Community District	Rent ($/sqft)	Vacancy (%)	Total # Storefronts	Total # Leases
Financial District	35.21	5.64	1433	645
Lower West Side	60.47	6.37	3740	1778
Lower East Side	31.68	5.39	3381	406
Midtown West	39.16	5.08	2207	658
Midtown	69.26	5.32	3954	2565
Midtown East	38.48	3.77	1918	533
Upper West Side	54.99	4.76	1888	505
Upper East Side	101.80	4.67	3290	901
Weighted Average	60.88	5.23		
Total			21811	7991

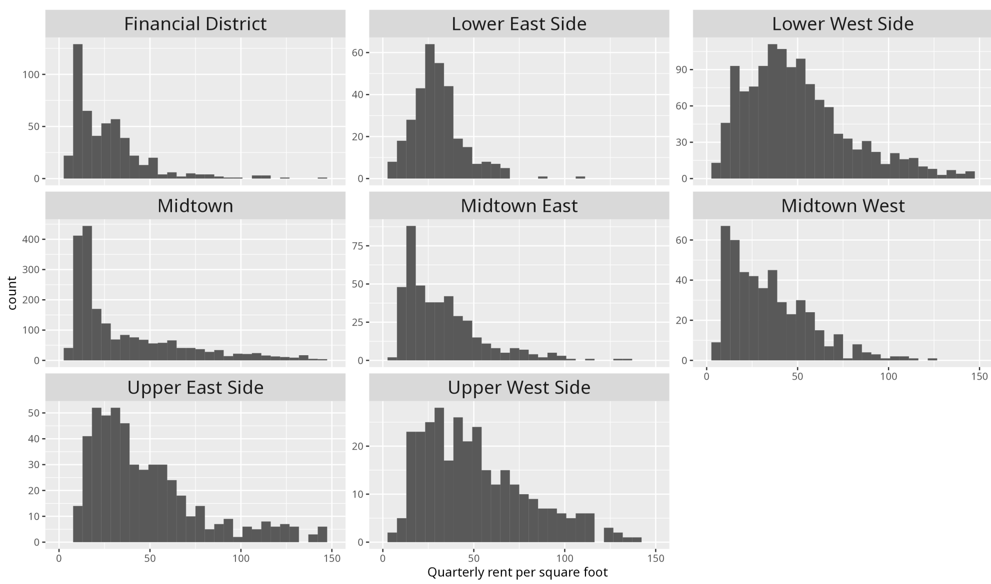
Note: We report summary statistics by neighborhood. From the CompStak dataset, we report the average quarterly rent per square foot and the total number of leases we observe from each market. Vacancy rates for 2007Q1-2017Q1 period come from the New York City Comptroller's report on retail vacancy; from 2017Q2 onward we compute vacancy rates from the Live XYZ dataset. The Comptroller's report provides vacancy rates at an annual frequency (corresponding to the first quarter of the year); we linearly interpolate to fill in quarterly vacancy rates in missing quarters. We also report the number of unique storefronts and leases observed in each market. To compute the average rent across neighborhoods, we weight each neighborhood's average rent by the number of leases we observe in that market. Similarly, we weight market-level vacancy rates by the number of storefronts in each market.

Figure 4: CompStak Leases Executed Per Quarter



Note: We plot the number of leases in the CompStak dataset that were executed in each quarter. The vertical line indicates the date CompStak was founded. CompStak's dataset is composed of transactions reported by commercial real estate brokers. Brokers are incentivized to report transactions because sharing information allows them to learn more about transactions they were not involved with. However, we want to be wary of leases reported prior to CompStak's entry, since brokers who report these transactions may be selected.

Figure 5: Rent Distribution By Community District



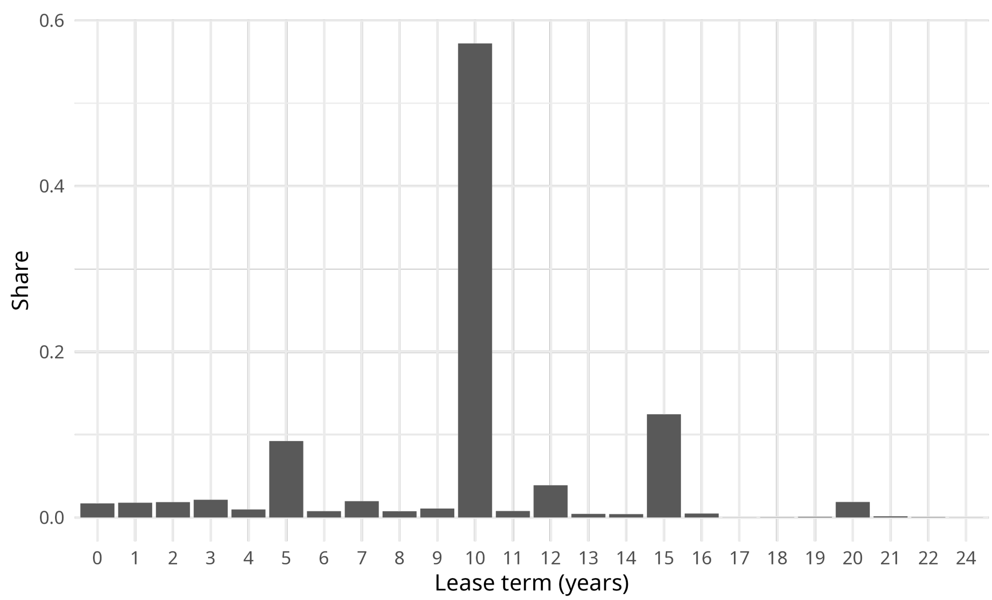
Note: We plot the distribution of real quarterly net effective rents (expressed in dollars per square foot) for each market from CompStak. Each observation is a lease, and we pool all leases observed over our whole 2005-2019 sample period.

Table 2: Correlates of Rent

Dependent Variable	(1) log(rent/sqft)	(2) log(rent/sqft)	(3) log(rent/sqft)	(4) log(rent/sqft)
Log Transaction Sqft	-0.20***	-0.22***	-0.23***	-0.23***
	(0.02)	(0.01)	(0.02)	(0.02)
Log Median Income	0.22***	0.14***	0.12***	-0.05
	(0.04)	(0.03)	(0.03)	(0.05)
Log Years Since Built	0.08**	0.02	0.001	-0.02
	(0.02)	(0.02)	(0.02)	(0.02)
Avenue Address	0.23***	0.24***	0.23***	0.24***
	(0.04)	(0.03)	(0.03)	(0.04)
Log Building Frontage	-0.06	-0.06*	-0.06**	-0.03
	(0.03)	(0.03)	(0.02)	(0.03)
Log Building Floors	0.17***	0.14***	0.06	0.02
	(0.04)	(0.03)	(0.03)	(0.04)
Residential Share	-0.48***	-0.33***	-0.13	-0.07
	(0.09)	(0.08)	(0.07)	(0.06)
Office Share	-0.38***	-0.20*	-0.11	-0.07
	(0.10)	(0.08)	(0.08)	(0.07)
Special Purpose District	0.16***	0.19***	0.03	0.13
	(0.03)	(0.03)	(0.04)	(0.07)
Chain	0.36***	0.24***	0.24***	0.21***
	(0.04)	(0.03)	(0.03)	(0.03)
Lease Term Years	0.0005	0.02***	0.02***	0.02***
	(0.005)	(0.005)	(0.004)	(0.004)
Fixed Effects				
Transaction Quarter	Yes	Yes	Yes	Yes
Tenant Industry	No	Yes	Yes	Yes
Zoning	No	No	Yes	Yes
Census Tract	No	No	No	Yes
Observations	2,630	2,630	2,630	2,630
R2	0.22518	0.40147	0.52022	0.62460
Within R2	0.20193	0.20548	0.14973	0.15012

Note: We regress log monthly rent per square foot on observable landlord and tenant characteristics using OLS. Our sample is the set of leases from the CompStak datast that we can match to tenants active in the Live XYZ dataset, so that we can include industry fixed effects and a dummy for whether the door is on a street (a small east-west side street) or an avenue (a major north-south thoroughfare). Standard errors are clustered at the transaction quarter and census tract level. Transaction square footage refers to the total amount of space being rented by the tenant. Median income is the income of the census tract in 2016 from the American Community Survey. Residential share and office share refer to the share of the entire building floor space devoted to residential and office uses, respectively, as reported by New York City's Primary Land Use Tax Lot Output (PLUTO) dataset. Special purpose districts have additional, unique zoning rules that vary on a case-by-case basis.

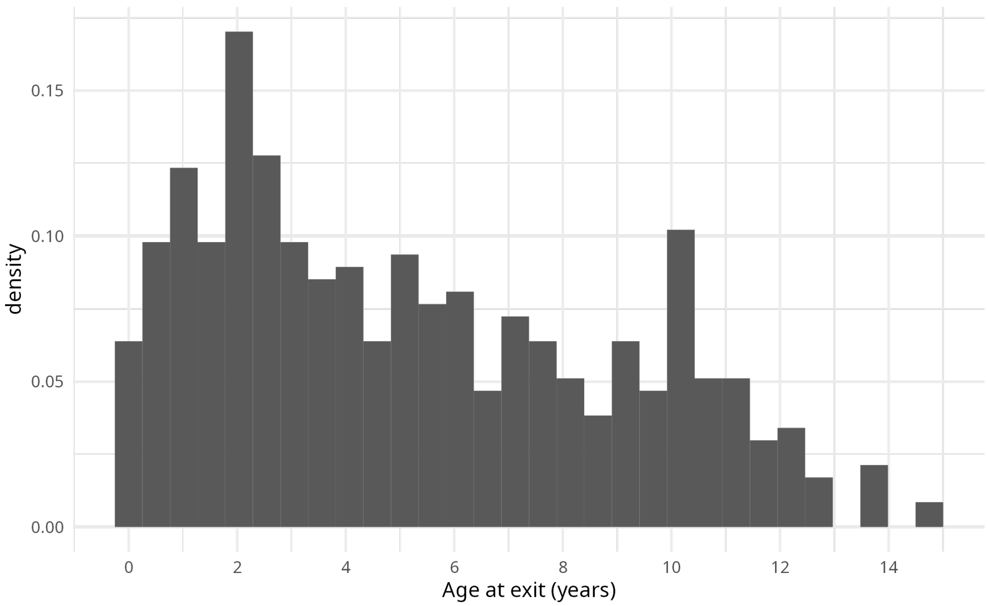
Figure 6: Contractual Lease Term Distribution



Note:

Figure 7: Lease Age at Exit

(a) Empirical Distribution



(b) Empirical Cumulative Distribution

Note: These figures are constructed from the sample of 462 tenants for whom we are able to match their lease in CompStak to their exit date in the Live XYZ dataset.

Figure 8: Aggregate State Path

Table 3: Aggregate State Transition Parameter Estimates

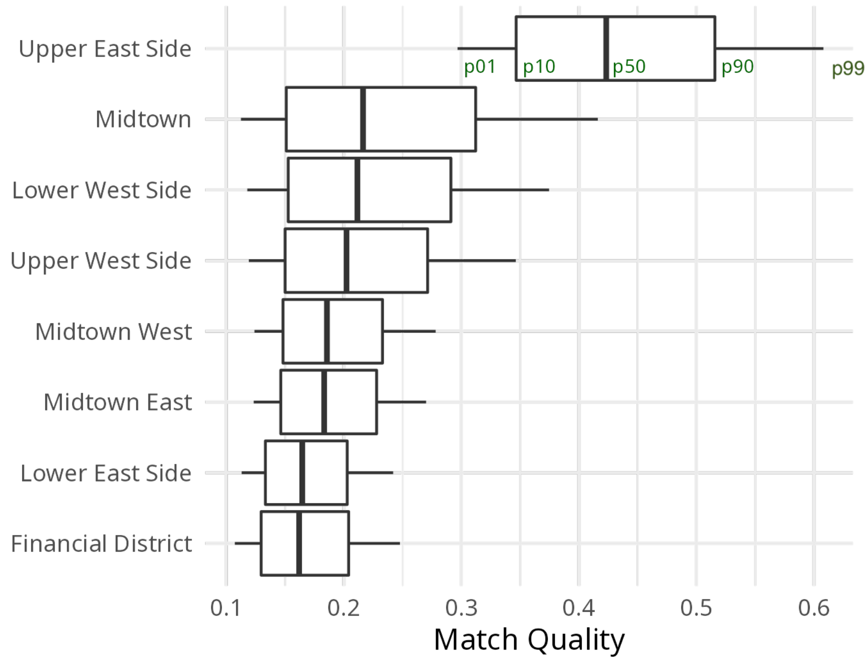
Parameter	Estimate
ρ_g	0.95 (0.04)
μ_g	333.75 (6.65)
σ_g^2	6.8

Table 4: Model Parameter Estimates

Community District	σ_ϕ	μ_θ	σ_θ	λ_0	λ_g
Financial District	2.21 (0.414)	-1.818 (0.002)	0.178 (0.003)	10.514 (5.222)	0.184 (0.003)
Lower West Side	2.185 (0.398)	-1.549 (0.01)	0.25 (0.007)	2.84 (0.551)	0.073 (0.009)
Lower East Side	2.433 (0.637)	-1.803 (0.011)	0.163 (0.005)	6.273 (0.651)	0.286 (0.01)
Midtown West	1.867 (0.169)	-1.682 (0.047)	0.174 (0.042)	15.635 (7.671)	0.131 (0.021)
Midtown	1.923 (0.331)	-1.528 (0.011)	0.281 (0.008)	8.175 (3.605)	-0.047 (0.032)
Midtown East	1.965 (0.099)	-1.697 (0.032)	0.17 (0.002)	21.035 (2.889)	0.249 (0.11)
Upper West Side	2.268 (0.443)	-1.596 (0.023)	0.232 (0.003)	12.092 (0.446)	0.17 (0.028)
Upper East Side	3.184 (2.115)	-0.859 (0.095)	0.156 (0.071)	-0.782 (0.628)	-0.157 (0.043)

Estimated parameters for each of our markets. μ_θ and σ_θ are the parameters of the lognormal distribution from which tenant types are drawn. σ_ϕ governs the mean of tenant opportunity costs; a tenant of age j has an average opportunity cost of $\sigma_\phi \times (T - j)$. Move-in costs m are calibrated to \$650 per square foot. λ_0 and λ_g are parameters of the matching function. Standard errors (in parentheses) are computed using a discrete approximation to the gradient of the moment condition.

Figure 9: Estimated Match Quality Distributions



For each nieghborhood, the thick vertical line represents the 50th percentile of the match quality distribution. The thin vertical lines represent the 10th and 90th percentiles, and the whiskers represent the first and 99th percentiles.

Figure 10: Effects of Reducing Variance

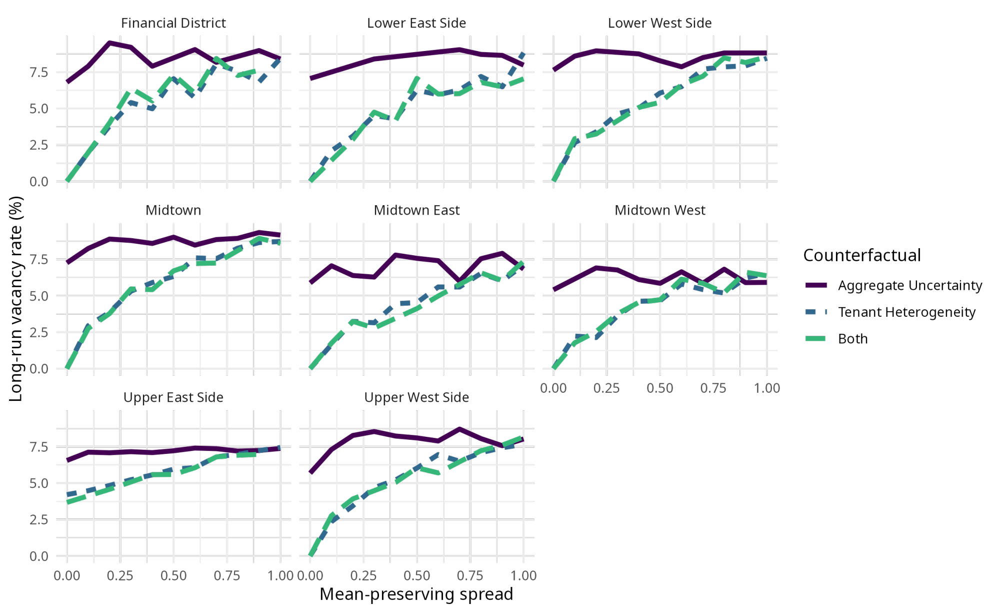


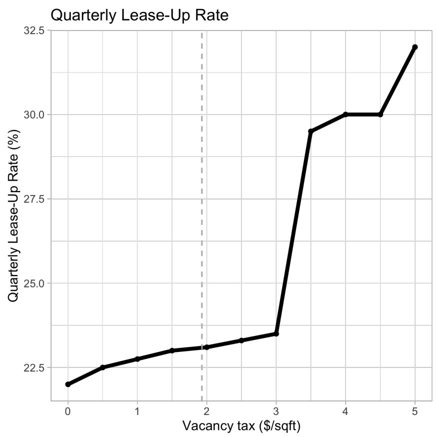
Table 5: Effect of Vacancy Tax on Long-Run Moments

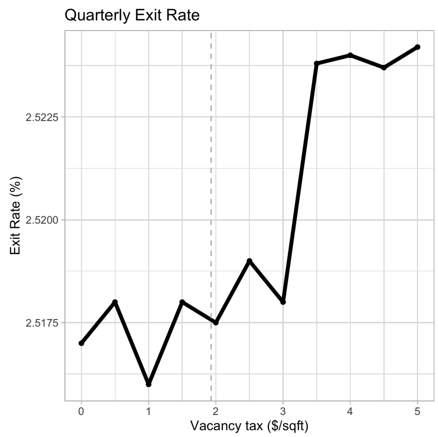
Community District	Vacancy Tax τ (\$/sqft)	Δ Vacancy Rate (%)	Δ Average Rent (%)	Implied Externality (per vacant sqft)
Financial District	0.98	-0.92	0.00	13.23
Lower West Side	1.93	-1.09	-0.16	12.26
Lower East Side	0.96	-0.98	-0.23	2.14
Midtown West	1.29	-0.61	0.00	20.33
Midtown	2.17	-3.05	-0.50	36.20
Midtown East	1.24	-1.71	-0.25	15.00
Upper West Side	1.74	-1.51	-0.18	5.00
Upper East Side	3.48	-13.25	-2.20	45.56
Average	1.72	-2.89	-0.44	18.72

This table reports the effects of a 1% tax on vacant assessed values for each of our 8 neighborhoods. We proxy for assessed value of vacant properties using the value of search in the model, $\mathbb{E}[U(g)]$. The levels of tax and the implied externality of vacancy are reported in dollars per square foot at a quarterly level. Relative to the estimated model, we report the percentage change in the long-run vacancy rate and in average rents.

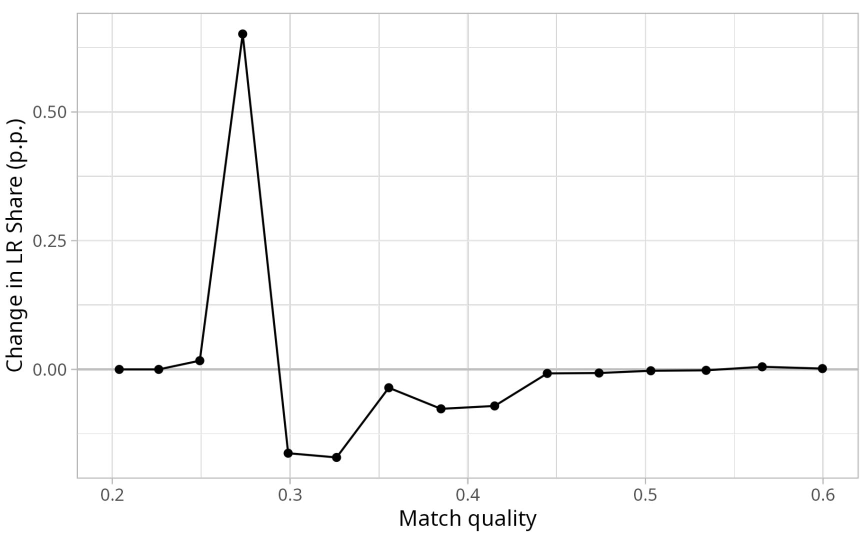
Figure 11: Effect of Vacancy Tax on Churn and Crowd-Out

(a) Establishment Churn





(b) Crowd-Out



These figures show the effect of increasing the vacancy tax on quarterly lease-up and exit rates on the Lower West Side. Results for other neighborhoods are similar. Large discrete jumps in both probabilities occur because we solve a discrete approximation to the continuous model presented in section 4.

Figure 12: Optimal Tax with No Vacancy Externalities

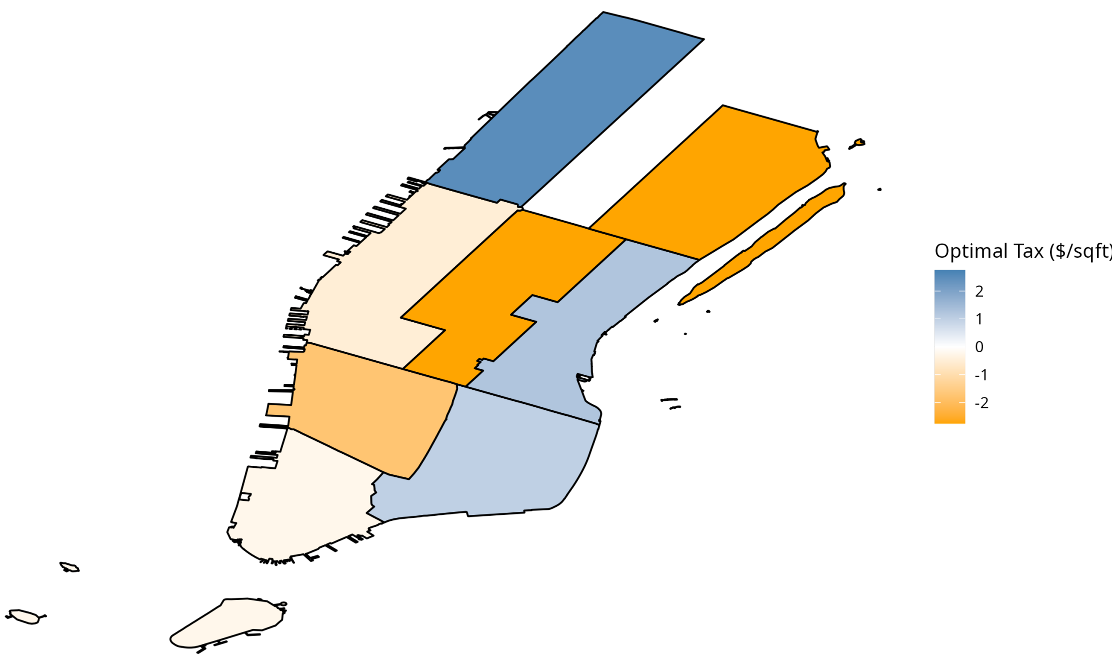


Figure 13: We map the optimal tax policy (in dollars per square foot) by community district in the absence of vacancy externalities ($e = 0$ in equation 24).

A Microfounding Tenant Profits

We assume a representative consumer with CES utility across the retail tenants i within a market. Each retailer offers a unique variety of retail goods or services:

$$U = \Big(\sum_i \alpha_i^{\frac{1}{\sigma}} x_i^{\frac{\sigma-1}{\sigma}}\Big)^{\frac{\sigma}{1-\sigma}} \text{subject to } \sum_i p_i x_i = B \tag{26}$$

where x_i is the quantity of variety i, p_i is the price of variety i, σ is the elasticity of substitution, B is the budget constraint, and α_i is a preference parameter for variety i.

We derive consumer demand in the usual way and obtain the standard demand curve:

$$x_a^{demand} = \frac{B p_a^{-\sigma}}{\sum_i \frac{\alpha_i}{\alpha_a} p_i^{1-\sigma}} \tag{27}$$

Tenants engage in Cournot competition. Specifically, the tenant producing variety a chooses a quantity x_a to produce to maximize static profits, given marginal costs c_a and a vector x_{-a} of quantities produced by all other varieties:

$$\pi_a(x_a, x_{-a}) = \max_{x_a} x_a \cdot (p_a(x_a, x_{-a}) - c_a) \tag{28}$$

The resulting supply curve is

$$x_a^{supply} = \left(\frac{B}{\sum_i \frac{\alpha_i}{\alpha_a} p_i^{1-\sigma}}\right)\left(\frac{1}{\sigma^\sigma (p_a - c_a)^\sigma}\right) \tag{29}$$

Setting demand (27) and supply (29) equal to each other allows us to solve for equilibrium prices and quantities. We obtain the familiar Cournot markup formula

$$p_a^* = \frac{\sigma}{\sigma - 1} c_a \tag{30}$$

and plug this into the supply curve to get equilibrium quantities:

$$x_a^* = \frac{B}{(\frac{\sigma}{\sigma-1})c_a + P_{-a}(\frac{\sigma}{\sigma-1})^\sigma c_a^\sigma} \tag{31}$$

Finally, we solve for equilibrium static flow profits by plugging equilibrium prices and quantities into the profit function for variety a:

$$\pi_a^* = \frac{B}{\sigma + A c_a^{\sigma-1}} \tag{32}$$

where $A = \sigma P_{-a} \sigma^{1-\sigma}(\sigma-1)^{1-\sigma}$.

In the structural leasing model of section 4, tenant quality θ_a represents $\frac{1}{\sigma+Ac_a^{\sigma-1}}$. The aggregate state variable g in the structural model corresponds to consumer budgets B. We therefore assume in the structural model that tenant profits take a multiplicative form: $\pi(g,\theta) = g\theta$.

B Model Proofs

Claim: $W(j,r,g,\phi;\theta)$ is strictly decreasing in r for all j,g,ϕ,θ.

Proof. Consider two arbitrary rent values, r and r', such that $r' > r$.

For lease age $j = T$, we have

$$W(T,r,g,\phi;\theta) = \pi(g,\theta) - r > \pi(g,\theta) - r' = W(T,r',g,\phi;\theta)$$

for all g,θ,ϕ.

Suppose for lease age $j+1$, $W(j,r,g,\phi;\theta) > W(j,r',g,\phi;\theta)$ for $r' > r$ and all g,θ,ϕ.

Then taking expectations over ϕ and g, we get

$$\begin{aligned}
\mathbb{E}_{\phi',g'}[W(j+1,r,g',\phi',\theta) \mid g] &= \int_{g',\phi'} W(j+1,r,g',\phi';\theta)dF(\phi',g' \mid g) \\
&> \int_{g',\phi'} W(j+1,r',g',\phi';\theta)dF(\phi',g' \mid g) \\
&= \mathbb{E}_{\phi',g'}[W(j+1,r',g',\phi',\theta) \mid g]
\end{aligned}$$

We now back up to compare the conditional values of exiting and staying in the lease in period j at rents r and r'. Since $-r > -r'$ and $\mathbb{E}_{\phi',g'}[W(j+1,r,g',\phi',\theta) \mid g] > \mathbb{E}_{\phi',g'}[W(j+1,r',g',\phi',\theta) \mid g]$, we have

$$
\begin{aligned}
W^{continue}(j, r, g; \theta) &= \pi(g, \theta) - r + \beta \mathbb{E}_{\phi', g'}[W(j+1, r, g', \phi', \theta) \mid g] \\
&> \pi(g, \theta) - r' + \beta \mathbb{E}_{\phi', g'}[W(j+1, r', g', \phi', \theta) \mid g] \\
&= W^{continue}(j, r', g; \theta)
\end{aligned}
$$

and

$$
W^{exit}(j, r, g, \phi; \theta) = -r + \phi > -r' + \phi = W^{exit}(j, r, g, \phi; \theta)
$$

Since $W^{continue}(j, r, g; \theta) > W^{continue}(j, r', g; \theta)$ and $W^{exit}(j, r, g, \phi; \theta) > W^{exit}(j, r', g, \phi; \theta)$,

$$
\begin{aligned}
W(j, r, g, \phi; \theta) &= \max\{W^{continue}(j, r, g; \theta), W^{exit}(j, r, g, \phi; \theta)\} \\
&> \max\{W^{continue}(j, r', g; \theta), W^{exit}(j, r', g, \phi; \theta)\} \\
&= W(j, r', g, \phi; \theta)
\end{aligned}
$$

□

Claim: $\phi^*(j, r, g; \theta)$ is strictly decreasing in r and $p^x(j, r, g; \theta)$ is strictly increasing in r for all $j < T, g, \phi, \theta$.

Proof. We know from the previous proof that $\mathbb{E}_{g,\phi}[W(j, r, g, \phi; \theta)] > \mathbb{E}_{g,\phi}[W(j, r', g, \phi; \theta)]$ for $r' > r$. Therefore

$$
\begin{aligned}
\phi^*(j, r', g; \theta) &= \pi(g, \theta) + \beta \mathbb{E}_{g,\phi}[W(j+1, r', g, \phi; \theta)] \\
&< \pi(g, \theta) + \beta \mathbb{E}_{g,\phi}[W(j+1, r, g, \phi; \theta)] = \phi^*(j, r, g, \theta)
\end{aligned}
$$

So $\phi^*(j, r, g, \theta)$ is strictly decreasing in r. As long as the CDF of ϕ is strictly increasing, then the tenant's exit probability $p^x(j, r, g; \theta)$ is increasing in r for all j, g, θ.

□

Claim: $W(j, r, g, \phi; \theta)$ is weakly increasing in θ for all $j, r, \phi, g > 0$.

Proof. Consider two different tenant qualities θ and θ' such that $\theta' > \theta$.

In the terminal lease period

$$
\begin{aligned}
W(T,r,g,\phi;\theta') &= \pi(g,\theta') - r \\
&= g\theta' - r \\
&> g\theta - r \\
&= \pi(g,\theta) - r \\
&= W(T,r,g,\phi;\theta)
\end{aligned}
$$

Suppose for lease age $j+1$ we know that $W(j+1,r,g,\phi;\theta') \geq W(j+1,r,g,\phi;\theta)$ for $\theta' > \theta$. Then taking expectations over ϕ and g, we get

$$
\begin{aligned}
\mathbb{E}_{\phi',g'}[W(j+1,r,g',\phi',\theta') \mid g] &= \int_{g',\phi'} W(j+1,r,g',\phi';\theta')dF(\phi',g' \mid g) \\
&\geq \int_{g',\phi'} W(j+1,r,g',\phi';\theta)dF(\phi',g' \mid g) \\
&= \mathbb{E}_{\phi',g'}[W(j+1,r,g',\phi',\theta) \mid g]
\end{aligned}
$$

We now back up to compare the conditional values of exiting and staying in the lease in period j for tenants of types θ and θ'. W^{exit} doesn't depend on θ, but $W^{continue}$ does and is strictly increasing in θ since $\pi(g,\theta)$ is strictly increasing in θ:

$$
\begin{aligned}
W^{continue}(j,r,g;\theta') &= \pi(g,\theta') - r + \beta\mathbb{E}_{\phi',g'}[W(j+1,r,g',\phi',\theta') \mid g] \\
&> \pi(g,\theta) - r + \beta\mathbb{E}_{\phi',g'}[W(j+1,r,g',\phi',\theta) \mid g] \\
&= W^{continue}(j,r,g;\theta)
\end{aligned}
$$

Since $W^{continue}(j,r,g;\theta') > W^{continue}(j,r,g;\theta)$ and $W^{exit}(j,r,g,\phi;\theta') = W^{exit}(j,r,g,\phi;\theta)$,

$$
\begin{aligned}
W(j,r,g,\phi;\theta') &= \max\{W^{continue}(j,r,g;\theta'), W^{exit}(j,r,g,\phi;\theta')\} \\
&\geq \max\{W^{continue}(j,r,g;\theta), W^{exit}(j,r,g,\phi;\theta)\} \\
&= W(j,r,g,\phi;\theta')
\end{aligned}
$$

□

Claim: $\phi^*(j,r,g;\theta)$ is strictly increasing and $p^x(j,r,g;\theta)$ is strictly decreasing in θ for all $j < T, r, \phi, g > 0$.

Proof. We know from the previous proof that $\mathbb{E}_{g,\phi}[W(j, r, g, \phi; \theta')] \geq \mathbb{E}_{g,\phi}[W(j, r, g, \phi; \theta)]$ for $\theta' > \theta$. We also know that for $g > 0$, $\pi(g, \theta) > \pi(g, \theta')$. Therefore:

$$\begin{aligned} \phi^*(j, r, g; \theta) &= \pi(g, \theta) + \beta \mathbb{E}_{g,\phi}[W(j + 1, r, g, \phi; \theta)] \\ &< \pi(g, \theta') + \beta \mathbb{E}_{g,\phi}[W(j + 1, r, g, \phi; \theta')] = \phi^*(j, r, g, \theta') \end{aligned}$$

So $\phi^*(j, r, g, \theta)$ is strictly decreasing in θ. As long as the CDF of ϕ is strictly increasing, then the tenant's exit probability $p^x(j, r, g; \theta)$ is decreasing in θ for all j, g, r.

$\square$